

US Patent & Trademark Office

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United States Patent Application Publication

20250268770

Kind Code

A1

Publication Date

August 28, 2025

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PATIENT SUPPORT SURFACE HAVING A PNEUMATIC CONTROL ASSEMBLY

Abstract

A pneumatic control assembly for a patient support surface includes a first air supply device providing air at a high flow rate and a low pressure, and a second air supply device providing air at a low flow rate and a high pressure. At least one first valve is configured to be fluidly connected to at least one inflatable section of the patient support surface, the at least one first valve being selectively actuated to fluidly connect the first air supply device with a corresponding one of the at least one inflatable section. The at least one first valve is piloted by the second air supply device. Other aspects of the pneumatic control assembly are also disclosed.

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Family ID: 1000008494774

Appl. No.: 19/063416

Filed: February 26, 2025

Related U.S. Application Data

us-provisional-application US 63558342 20240227

Publication Classification

Int. Cl.: A61G7/057 (20060101); A61G7/075 (20060101)

Background/Summary

CROSS-REFERENCE [0001] The present application claims priority from U.S. Provisional Patent Application No. 63/558,342, filed Feb. 27, 2024, the entirety of which is incorporated by reference herein.

FIELD OF TECHNOLOGY

[0002] The present technology relates to patient support surfaces such as mattresses for patient support apparatuses such as hospital beds, and particularly to inflatable patient support surfaces.

BACKGROUND

[0003] Inflatable mattresses are typically used on patient support apparatuses, namely hospital beds, to offer adaptable support for patients and different types of functions and therapies that are not easily achievable with a non-inflatable mattress. For instance, such inflatable mattresses may be available in more demanding environments such as in an intensive care unit (ICU) of a hospital. These inflatable mattresses usually include a pneumatic control assembly (sometimes referred to as a “pneumatic box” or “valve control box”) enclosed within the mattress to control the inflation of various inflatable bladders of the mattress. In some cases, the pneumatic control assembly may also be disposed outside of the mattress (e.g., in a separate control module disposed on the bed).

[0004] The design of the pneumatic control assembly poses various challenges. In particular, the pneumatic control assembly typically includes a multitude of valves to control air flow to the different inflatable bladders of the mattress in a relatively restricted volume defined by a housing of the pneumatic control assembly. The valves also usually produce a significant amount of heat during operation which is undesirable for the performance of the electric components of the control assembly and/or for heat management of the mattress itself. Furthermore, it is desirable to minimize the noise emitted by the pneumatic control assembly, namely by the valves thereof and/or by a fluid source (e.g., a blower).

[0005] In view of the foregoing, there is a need for a patient support surface that addresses at least some of these drawbacks.

SUMMARY

[0006] It is an object of the present technology to ameliorate at least some of the inconveniences present in the prior art.

[0007] According to an aspect of the present technology, there is provided a pneumatic control assembly for a patient support surface, comprising: a first air supply device providing air at a high flow rate and a low pressure; a second air supply device providing air at a low flow rate and a high pressure; and at least one first valve configured to be fluidly connected to at least one inflatable section of the patient support surface, the at least one first valve being selectively actuated to fluidly connect the first air supply device with a corresponding one of the at least one inflatable section, the at least one first valve being piloted by the second air supply device.

[0008] In some embodiments, the first air supply device provides air at a first flow rate and a first pressure; the second air supply device provides air at a second flow rate and a second pressure; the first flow rate is greater than the second flow rate; and the second pressure is greater than the first pressure.

[0009] In some embodiments, the first air supply device is a blower and the second air supply device is a compressor.

[0010] In some embodiments, the first air supply device has a pressure capacity that is less than

kPa.

[0011] In some embodiments, the second air supply device has a pressure capacity that is more than 50 kPa.

[0012] In some embodiments, each of the at least one first valve has a pilot port; the pneumatic control assembly further comprises at least one solenoid valve, the at least one solenoid valve being fluidly connected to the second air supply device and to the pilot port of the at least one first valve; and the at least one solenoid valve is selectively actuated to control air flow from the second air supply device to the pilot port of the at least one first valve to thereby actuate the at least one first valve.

[0013] In some embodiments, each of the at least one first valve is a diaphragm valve.

[0014] In some embodiments, the pneumatic control assembly further comprises a manifold comprising the at least one first valve, the manifold being in fluid communication with the first air supply device.

[0015] In some embodiments, the at least one inflatable section is at least one first inflatable section; the first air supply device is in selective fluid communication with the at least one first inflatable section via at least one first valve; the pneumatic control assembly further comprises: a third air supply device providing air at the high flow rate and the low pressure; and at least one second valve configured to be fluidly connected to at least one second inflatable section of the patient support surface, the at least one second valve being selectively actuated to fluidly connect the third air supply device with the at least one second inflatable section, the at least one second valve being piloted by the second air supply device.

[0016] In some embodiments, the third air supply device is a blower.

[0017] In some embodiments, the pneumatic control assembly further comprises: a reservoir that is fluidly connected to the second air supply device, the reservoir being filled by the second air supply device to contain pressurized air therein, the at least one first valve being selectively piloted by the reservoir.

[0018] In some embodiments, the pneumatic control assembly further comprises a housing enclosing the first air supply device, the second air supply device and the at least one first valve.

[0019] In some embodiments, a patient support surface for a patient support apparatus comprises: a plurality of inflatable sections; the pneumatic control assembly for controlling air flow to the inflatable sections; and a cover enclosing the inflatable sections and the pneumatic control assembly.

[0020] In some embodiments, the pneumatic control assembly is located near a foot end of the patient support surface.

[0021] In some embodiments, the inflatable sections include an inflatable foot section, the pneumatic control assembly being disposed underneath the inflatable foot section.

[0022] According to another aspect of the present technology, there is provided a pneumatic control assembly for a patient support surface, comprising: a first air supply device fluidly connected to a first inlet; at least one first valve fluidly connected to the first air supply device, the at least one first valve being configured to be fluidly connected to a respective one of at least one first inflatable section of the patient support surface; a second air supply device fluidly connected to a second inlet; and at least one second valve fluidly connected to the second air supply device, the at least one second valve being configured to be fluidly connected to a respective one of at least one second inflatable section of the patient support surface, the pneumatic control assembly being operable in: a direct feed mode wherein (i) the first air supply device pumps air from the first inlet to the at least one first valve, and/or (ii) the second air supply device pumps air from the second inlet to the at least one second valve; and a dual-stage compression mode wherein a given one of the first air supply device and the second air supply device pumps air from a corresponding one of the first inlet and the second inlet to an other one of the first air supply device and the second air supply device, and the other one of the first air supply device and the second air supply device

pumps air received from the given one of the first air supply device and the second air supply device to a corresponding one of the at least one first valve and the at least one second valve.

[0023] In some embodiments, in the dual-stage compression mode: the given one of the first air supply device and the second air supply device is the first air supply device in response to a request to increase a pressure associated with the at least one second inflatable section; and the given one of the first air supply device and the second air supply device is the second air supply device in response to a request to increase a pressure associated with the at least one first inflatable section.

[0024] In some embodiments, the direct feed mode is engaged in response to a requested target pressure associated with the at least one first inflatable section or the at least second inflatable section is less than a threshold pressure; and the dual-stage compression mode is engaged in response to the requested target pressure associated with the at least one first inflatable section or the at least one second inflatable section is greater than the threshold pressure.

[0025] In some embodiments, the threshold pressure is less than a pressure capacity of each of the first and second air supply devices.

[0026] In some embodiments, each of the first air supply device and the second air supply device is a blower.

[0027] In some embodiments, each of the first air supply device and the second air supply device has a pressure capacity of between 4 and 6 kPa inclusively.

[0028] In some embodiments, the pneumatic control assembly further comprises: a first interconnection valve selectively fluidly connecting an outlet of the second air supply device with an inlet of the first air supply device; and a second interconnection valve selectively fluidly connecting an outlet of the first air supply device with an inlet of the second air supply device. In the dual-stage compression mode, one of the first interconnection valve and the second interconnection valve is opened, and an other one of the first interconnection valve and the second interconnection valve remains closed.

[0029] In some embodiments, the first and second interconnection valves are air piloted.

[0030] In some embodiments, the pneumatic control assembly further comprises: a first solenoid valve controlling air flow to a pilot port of the first interconnection valve; and a second solenoid valve controlling air flow to a pilot port of the second interconnection valve, the first and second solenoid valves being actuated to cause the first and second interconnection valves to be actuated respectively.

[0031] In some embodiments, the first and second interconnection valves are diaphragm valves.

[0032] In some embodiments, the pneumatic control assembly further comprises: a first manifold comprising the at least one first valve and the first interconnection valve; and a second manifold comprising the at least one second valve and the second interconnection valve.

[0033] In some embodiments, each of the first manifold and the second manifold comprises a main body defining at least in part a main valve chamber and an interconnection channel; the at least one first valve is disposed in the main valve chamber of the first manifold; the at least one second valve is disposed in the main valve chamber of the second manifold; the interconnection channel of the first manifold is fluidly connected to the main valve chamber of the second manifold; and the interconnection channel of the second manifold is fluidly connected to the main valve chamber of the first manifold.

[0034] In some embodiments, the main body of each of the first manifold and the second manifold defines at least in part a secondary valve chamber sealed off from the main valve chamber; the first interconnection valve is disposed in the secondary valve chamber of the first manifold; and the second interconnection valve is disposed in the secondary valve chamber of the second manifold.

[0035] In some embodiments, each of the first manifold and the second manifold: defines a first aperture opening into the interconnection channel; and a second aperture opening into the main valve chamber; the first aperture of the first manifold is fluidly connected to the second aperture of the second manifold; and the second aperture of the first manifold is fluidly connected to the first

aperture of the first manifold.

[0036] In some embodiments, for each of the first manifold and the second manifold, the first and second apertures are located at a lateral end of the manifold; and the first manifold and the second manifold are oriented such that the first and second apertures of the first manifold are disposed near the first and second apertures of the second manifold.

[0037] In some embodiments, the first manifold and the second manifold are identical.

[0038] In some embodiments, the pneumatic control assembly is also operable in a deflate mode to depressurize the at least one first inflatable section or the at least one second inflatable section; in the deflate mode: (i) if the at least one first inflatable section is to be depressurized: the second interconnection valve is opened to fluidly connect the inlet of the second air supply device to the main valve chamber of the first manifold; a second inlet valve is closed to fluidly disconnect the second air supply device from the second inlet; the at least one first valve is opened to fluidly connect the at least one first inflatable section of the patient support surface with the main valve chamber of the first manifold; and the second air supply device is activated to draw air from the at least one first valve and thus from the at least one first inflatable section associated therewith; and (ii) if the at least one second inflatable section is to be depressurized: the first interconnection valve is opened to fluidly connect the inlet of the first air supply device to the main valve chamber of the second manifold; a first inlet valve is closed to fluidly disconnect the first air supply device from the first inlet; the at least one second valve is opened to fluidly connect the at least one second inflatable section of the patient support surface with the main valve chamber of the second manifold; and the first air supply device is activated to draw air from the at least one second valve and thus from the at least one second inflatable section associated therewith.

[0039] In some embodiments, the pneumatic control assembly further comprises a discharge valve fluidly connected to at least one of the first air supply device and the second air supply device; the discharge valve is configured to be fluidly connected to one of an internal space of the patient support surface and a component of the patient support surface other than an inflatable section; in the deflate mode: if the at least one first inflatable section is to be depressurized, the discharge valve is opened such that the second air supply device pumps air drawn from the at least one first valve through the discharge valve; and if the at least one second inflatable section is to be depressurized, the discharge valve is opened such that the second air supply device pumps air drawn from the at least one second valve through the discharge valve.

[0040] In some embodiments, the discharge valve is associated with a low air loss function of the patient support surface.

[0041] In some embodiments, in the deflate mode: if the at least one first inflatable section is to be depressurized, the first interconnection valve is opened and the first inlet valve is opened to fluidly communicate an outlet of the second air supply device to the first inlet valve in order to discharge air drawn from the at least one first valve through the first inlet valve; and if the at least one second inflatable section is to be depressurized, the second interconnection valve is opened and the second inlet valve is opened to fluidly communicate an outlet of the first air supply device to the second inlet valve in order to discharge air drawn from the at least one second valve through the second inlet valve.

[0042] In some embodiments, each of the first air supply device and the second air supply device provides air at a high flow rate and a low pressure; the pneumatic control assembly further comprises a third air supply device providing air at a low flow rate and a high pressure, the third air supply device being operable to pilot the at least one first valve and the at least one second valve in the direct feed mode and the dual-stage compression mode; the third air supply device is selectively fluidly connected to the at least one first valve and the at least one second valve; the pneumatic control assembly is also operable in a high pressure feed mode to pressurize the at least one first inflatable section or the at least one second inflatable section; in the high pressure feed mode, the third air supply device delivers air to the at least one first valve or the at least one second valve in

order to pressurize the at least one first inflatable section or the at least one second inflatable section.

[0043] In some embodiments, a patient support surface comprises: a plurality of inflatable sections for supporting a patient; the pneumatic control assembly; and a cover enclosing the inflatable sections and the pneumatic control assembly.

[0044] According to another aspect of the present technology, there is provided a patient support surface comprising: a plurality of inflatable sections for supporting a patient; and a pneumatic control assembly for controlling a pressure within the inflatable sections of the patient support surface, the pneumatic control assembly comprising a first air supply device and a second air supply device for inflating and deflating the inflatable sections, the pneumatic control assembly being operable in a direct feed mode and a dual-stage compression mode to pressurize a given one of the inflatable sections of the patient support surface, wherein: in the direct feed mode, the first air supply device draws air external to the pneumatic control assembly and drives said air to the given one of the inflatable sections while bypassing the second air supply device, and in the dual-stage compression mode, the first air supply device draws air external to the pneumatic control assembly and drives said air to the second air supply device, the second air supply device then driving said air to the given one of the inflatable sections.

[0045] In some embodiments, the pneumatic control assembly further comprises a controller in communication with the first air supply device and the second air supply device; the controller operates the pneumatic control assembly in the direct feed mode or in the dual-stage compression mode to increase a current pressure of the given one of the inflatable sections to a target pressure, the controller operating in: (i) the direct feed mode in response to the target pressure being less than a threshold pressure; and (ii) the dual-stage compression mode in response to the target pressure being greater than the threshold pressure.

[0046] In some embodiments, the threshold pressure is less than a pressure capacity of each of the first and second air supplies.

[0047] In some embodiments, the first air supply device and the second air supply device have a same pressure capacity and flow rate.

[0048] According to another aspect of the present technology, there is provided a method for controlling a pneumatic control assembly of a patient support surface, the pneumatic control assembly comprising a first air supply device and a second air supply device, the method comprising: determining a target pressure for a given inflatable section of the patient support surface; comparing the target pressure with a current pressure of the given inflatable section; in response to the target pressure being greater than the current pressure and less than a threshold pressure, engaging a direct feed mode comprising: drawing air external to the pneumatic control assembly with the first air supply device; and driving air flow, by the first air supply device, to the given inflatable section while bypassing the second air supply device; in response to the target pressure being greater than the current pressure and greater than the threshold pressure, engaging a dual-stage compression mode comprising: drawing air external to the pneumatic control assembly with the first air supply device; driving air flow, by the first air supply device, to the second air supply device; and after driving air flow to the second air supply device, driving air flow, by the second air supply device, to the given inflatable section.

[0049] In some embodiments, the threshold pressure is less than a pressure capacity of each of the first and second air supply devices.

[0050] In some embodiments, the first air supply device and the second air supply device have a same pressure capacity and flow rate.

[0051] In some embodiments, the method further comprises: in response to the target pressure being less than the current pressure, engaging a deflate mode comprising: drawing air from the given inflatable section by the first air supply device; and driving air flow, by the first air supply device, outside of a pneumatic system of the patient support surface.

[0052] In some embodiments, the pneumatic control assembly further comprises a third air supply device that provides air at a greater pressure and a lower flow rate than the first and second air supply devices; the method further comprises, in response to the target pressure being greater than the current pressure and a required pressure adjustment being less than a threshold pressure adjustment, engaging a high pressure feed mode comprising: drawing air external to the pneumatic control assembly by the third air supply device; and driving air flow, by the third air supply device, to the given inflatable section.

[0053] According to another aspect of the present technology, there is provided a pneumatic control assembly for a patient support surface, comprising: a first air supply device providing air at a high flow rate and a low pressure; a second air supply device providing air at a low flow rate and a high pressure; and at least one valve configured to be fluidly connected to at least one inflatable section of the patient support surface, in a first operation mode, the first air supply device delivers air to the at least one valve in order to pressurize the at least one inflatable section, and in a second operation mode, the second air supply device delivers air to the at least one valve in order to pressurize the at least one inflatable section.

[0054] In some embodiments, the second operation mode is engaged in response to a pressure adjustment required at the at least one inflatable section being less than a threshold pressure adjustment.

[0055] In some embodiments, the first operation mode is engaged in response to the pressure adjustment at the at least one inflatable section being greater than the threshold pressure adjustment.

[0056] In some embodiments, each of the at least one valve has a pilot port; the second air supply device is fluidly connected to the pilot port of each of the at least one valve for actuation thereof; and in the first operation mode, the at least one valve is piloted by the second air supply device so that, in response to being actuated by the second air supply device, the at least one valve discharges air flow from the first air supply device to pressurize the at least one inflatable section.

[0057] In some embodiments, the pneumatic control assembly further comprises a reservoir containing pressurized air, the reservoir being in fluid communication with the pilot port of the at least one valve in order to actuate the at least one valve in the second operation mode.

[0058] In some embodiments, the reservoir is in selective fluid communication with the second air supply device, the reservoir being filled by the second air supply device.

[0059] In some embodiments, the pneumatic control assembly further comprises: at least one solenoid valve configured to selectively fluidly communicate the second air supply device with the at least one valve, in the first operation mode, air flow discharged by the second air supply device is routed to the at least one valve via the at least one solenoid valve, and in the second operation mode, air flow discharged by the second air supply device reaches the at least one valve by bypassing the at least one solenoid valve.

[0060] In some embodiments, the pneumatic control assembly further comprises a controller in communication with and controlling the first air supply device, the second air supply device and the at least one solenoid valve.

[0061] In some embodiments, the first air supply device is deactivated in the second operation mode.

[0062] In some embodiments, the first air supply device provides air at a first flow rate and a first pressure; the second air supply device provides air at a second flow rate and a second pressure; the first flow rate is greater than second flow rate; and the second pressure is greater than the first pressure.

[0063] In some embodiments, the first air supply device is a blower and the second air supply device is a compressor.

[0064] In some embodiments, the first air supply device has a pressure capacity that is less than kPa.

[0065] In some embodiments, the second air supply device has a pressure capacity that is more than 50 kPa.

[0066] In some embodiments, each of the at least one valve is a diaphragm valve.

[0067] In some embodiments, the at least one valve is a plurality of valves; and the pneumatic control assembly further comprises a manifold comprising the plurality of valves.

[0068] In some embodiments, the pneumatic control assembly further comprises a housing enclosing the first air supply device, the second air supply device and the at least one valve.

[0069] In some embodiments, a patient support surface for a patient support apparatus comprises: a plurality of inflatable sections for supporting a patient; the pneumatic control assembly for controlling air flow to the inflatable sections; and a cover enclosing the inflatable sections and the pneumatic control assembly.

[0070] In some embodiments, the pneumatic control assembly is located near a foot end of the patient support surface.

[0071] In some embodiments, the inflatable sections include an inflatable foot section, the pneumatic control assembly being disposed underneath the inflatable foot section.

[0072] According to another aspect of the present technology there is provided a pneumatic control assembly for a patient support surface, comprising: a first air supply device; a second air supply device; and at least one diaphragm valve for selectively fluidly connecting the first air supply device with at least one air-powered part of the patient support surface, each of the at least one diaphragm valve comprising: a pilot port in selective fluid communication with the second air supply device; an outlet aperture configured to be fluidly connected with the at least one air-powered part of the patient support surface; and a diaphragm that is selectively deformable from a first position to a second position in response to the pilot port being in fluid communication with the second air supply device, in the first position, the diaphragm being clear of the outlet aperture to allow air flow from the first air supply device therethrough, in the second position, the diaphragm blocking the outlet aperture to prevent air flow from the first air supply device therethrough.

[0073] In some embodiments, the diaphragm is biased to be in the first position.

[0074] In some embodiments, the at least one diaphragm valve is a plurality of diaphragm valves; the pneumatic control assembly further comprises at least one manifold, each of the at least one manifold comprising: a main body defining at least in part an internal chamber, the internal chamber being in fluid communication with the first air supply device; and the plurality of diaphragm valves contained at least in part within the internal chamber.

[0075] In some embodiments, the at least one manifold comprises a first manifold and a second manifold, the second manifold being identical to the first manifold.

[0076] In some embodiments, the first manifold is oriented at a 180° offset relative to the second manifold.

[0077] In some embodiments, the at least one manifold comprises a first manifold and a second manifold; and the first manifold is in selective fluid communication with the second manifold.

[0078] According to another aspect of the present technology, there is provided a valve module for use in a pneumatic control assembly of a patient support surface, the valve module comprising: a first member comprising a pilot port; a diaphragm that is selectively deformable; and a second member coupled to the first member and retaining a peripheral portion of the diaphragm therebetween.

[0079] In some embodiments, the second member has a stepped peripheral portion.

[0080] In some embodiments, the valve module is generally disc-shaped.

[0081] In some embodiments, a chamber is formed between the first member and the diaphragm that is pressurized via the pilot port.

[0082] According to another aspect of the present technology, there is provided an air supply unit for a patient support surface, comprising: an enclosure; an air supply device contained within the

enclosure; and an attenuating filler extending between an outer surface of the air supply device and an inner surface of the enclosure to fix the air supply device in place within the enclosure.

[0083] In some embodiments, the attenuating filler comprises a plurality of filler members.

[0084] In some embodiments, the plurality of filler members includes an upper filler member overlying the air supply device and a lower filler member underlying the air supply device.

[0085] In some embodiments, the attenuating filler defines at least one recess that is a negative of a shape of the air supply device.

[0086] In some embodiments, the enclosure has an enclosure inlet and an enclosure outlet; the air supply device has a device inlet and a device outlet; and the attenuating filler defines at least one channel for fluidly connecting the enclosure inlet to the device inlet and at least one channel for fluidly connecting the enclosure outlet to the device outlet.

[0087] In some embodiments, the air supply unit further comprises a flexible sleeve connecting the device outlet to the enclosure outlet.

[0088] In some embodiments, the flexible sleeve is made of fabric.

[0089] In some embodiments, the air supply device is a blower.

[0090] In some embodiments, a pneumatic control assembly for the patient support surface comprises: a housing; the air supply unit enclosed within the housing, a chamber being defined between an inner surface of the enclosure of the air supply unit and the housing of the pneumatic control assembly, the air supply device being disposed within the chamber, the air supply device being detached from the housing of the pneumatic control assembly.

[0091] Embodiments of the present technology each have at least one of the above-mentioned objects and/or aspects, but do not necessarily have all of them. It should be understood that some aspects of the present technology that have resulted from attempting to attain the above-mentioned objects may not satisfy these objects and/or may satisfy other objects not specifically recited herein.

[0092] Additional and/or alternative features, aspects and advantages of embodiments of the present technology will become apparent from the following description, the accompanying drawings and the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0093] For a better understanding of the present technology, as well as other aspects and further features thereof, reference is made to the following description which is to be used in conjunction with the accompanying drawings, where:

[0094] FIG. 1 is a perspective view, taken from a top, left side, of a mattress according to an embodiment of the present technology, shown with a patient lying on the mattress;

[0095] FIG. 2 is a perspective view, taken from a top, right side, of a patient support apparatus configured to support the mattress of FIG. 1;

[0096] FIG. 3 is a right side elevation view of the patient support apparatus of FIG. 2, showing the patient support apparatus in a chair egress position;

[0097] FIG. 4 is a perspective view, taken from a top, left side of the mattress of FIG. 1, with a top portion of a cover of the mattress removed to expose the internal components of the mattress;

[0098] FIG. 5 is a top plan view of the mattress of FIG. 4;

[0099] FIG. 6A is a cross-sectional view of the mattress taken along line 6A-6A in FIG. 5;

[0100] FIG. 6B is a cross-sectional view of the mattress taken along the same plane as FIG. 6A, showing inflatable lateral extensions of the mattress in a collapsed state;

[0101] FIG. 7 is a perspective view, taken from atop, right side of a pneumatic control assembly of the mattress of FIG. 4, with a cover portion of a housing of the pneumatic control assembly being

shown removed from a base portion of the housing;

[0102] FIG. **8** is a top plan view of part of the pneumatic control assembly of FIG. **7**, including the base portion of the housing and internal components received thereby, shown with blower housings removed to expose respective blowers disposed therein;

[0103] FIG. **9** is a bottom plan view of the part of the pneumatic control assembly of FIG. **7**;

[0104] FIG. **10** is a flow diagram of a method for controlling the pneumatic control assembly in accordance with an embodiment of the present technology;

[0105] FIG. **11** is a schematic diagram of a pneumatic system of the pneumatic control assembly of FIG. **7**;

[0106] FIG. **12** is a top perspective view of a manifold of the pneumatic control assembly of FIG. **7**;

[0107] FIG. **13** is a bottom perspective view of the manifold of FIG. **12**;

[0108] FIG. **14** is an exploded view of the manifold of FIG. **12**;

[0109] FIG. **15** is a cross-sectional view taken along line **15-15** shown in FIG. **12**;

[0110] FIG. **16** is a perspective view, taken from a top, left side, of a main body of the manifold of FIG. **12**;

[0111] FIG. **17** is a perspective view, taken from bottom, left side, of the main body of FIG. **16**;

[0112] FIG. **18** is a perspective view, taken from a top, left side, of an outlet connector of the manifold of FIG. **12**;

[0113] FIG. **19** is a top plan view of the outlet connector of FIG. **18**;

[0114] FIG. **20** is a top perspective view of a valve module of the manifold of FIG. **12**;

[0115] FIG. **21** is a cross-sectional view of the valve module of FIG. **20** taken along line **21-21** in FIG. **20**;

[0116] FIG. **22** is an exploded view of the valve module of FIG. **20**;

[0117] FIG. **23** is a voltage to time graph representing an applied voltage of a compressor of the pneumatic control assembly according to one embodiment;

[0118] FIG. **24** is a perspective view, taken from top left side, of a valve module according to another embodiment;

[0119] FIG. **25** is a cross-sectional view of the valve module of FIG. **24** taken along line **25-25** in FIG. **24**;

[0120] FIG. **26** is an exploded view of the valve module of FIG. **24**;

[0121] FIG. **27** is a schematic diagram of an alternative embodiment of the pneumatic system of the pneumatic control assembly;

[0122] FIG. **28** is a top perspective view of two manifolds of the pneumatic control assembly according to an alternative embodiment;

[0123] FIG. **29** is a top perspective view of an air supply unit of the pneumatic control assembly;

[0124] FIG. **30** is a top plan view of the air supply unit of FIG. **29**;

[0125] FIG. **31** is an exploded view of the air supply unit of FIG. **29**; and

[0126] FIG. **32** is a cross-sectional view of the air supply unit of FIG. **29** taken along line **32-32** of FIG. **30**.

DETAILED DESCRIPTION

[0127] A mattress **100** in accordance with an embodiment of the present technology is illustrated in FIG. **1**. The mattress **100** may be used in a medical setting to comfortably support a patient thereon. The mattress **100** may alternatively be referred to as a “patient support surface”. As shown in FIGS. **2** and **3**, the mattress **100** can be used in conjunction with a patient support apparatus **300** such as a hospital bed. As will be explained in greater detail below, the mattress **100** is constituted in part by multiple inflatable bladders that are selectively pressurized to varying degrees to provide different features to the mattress **100**, such as different treatments, movement facilitation and/or compatibility with certain functions of the patient support apparatus **300** amongst others.

[0128] With reference to FIGS. **2** and **3**, in this embodiment, the patient support apparatus **300** is a

hospital bed intended for use in an intensive care unit (ICU). As such, the bed **300** has advanced functions that are typically not available in an ordinary hospital bed. Nevertheless, it is contemplated that, in other embodiments, the bed **300** may be an ordinary hospital bed that does not have all the advanced functions expected from an ICU bed. The bed **300** has a base **302**, an upper frame **304** operatively connected to the base **302**, and a deck **306** connected to the upper frame **304**. The deck **306** includes multiple deck sections **307**, **309**, **311**, **313** that are movable relative to each other and define a surface **308** of the deck **306** on which, in use, the mattress **100** is supported. The bed **300** also has a headboard **310** at a head end thereof and a footboard **312** at a foot end thereof. A plurality of siderails **314** is also provided on which one or more user interfaces (e.g., a graphic display such as a touch screen) may be provided to allow a user (e.g., a medical practitioner) to control certain functions of the bed **300** and/or the mattress **100** as well as to access information relating to the bed **300**, the mattress **100** and/or the patient.

[0129] An elevation system **316** operatively connects the upper frame **304** to the base **302** and allows adjustment of the height of the upper frame **304**. In this embodiment, the elevation system **316** includes two lift columns **318** that are longitudinally spaced from each other. Each lift column **318** includes an actuator (not shown) that is operable to adjust the height of the lift column **318**. The height of the lift columns **318** can be adjusted separately such that the head end and foot end of the upper frame **104** is set at different heights. As such, the bed **300** can be set to different positions including a flat horizontal position (shown in FIG. 2), also known as the bed position, a Trendelenburg position, a reverse Trendelenburg position, a cardiac chair position, and a chair egress position (shown in FIG. 3), amongst others.

[0130] The bed **300** also has an adjustable length, by selectively extending and retracting a foot deck section **313** of the deck **306**, and an adjustable width by selectively extending and retracting slidable panels from each side of one of the deck sections **307**, **309**, **311**, **313** of the deck **306**. Notably, in this embodiment, the bed **300** is designed to be used by bariatric patients which typically will need a bed with a larger width due to their size.

[0131] A more complete description of the construction and features of the bed **300** can be found in International Patent Application Publication No. 2023/002370A1, published on Jan. 26, 2023, the entirety of which is incorporated by reference herein.

[0132] The bed **300** may be configured differently in other embodiments. For instance, in some embodiments, the elevation system **316** may instead have a head end lift assembly and a foot end lift assembly, each having a generally inverted-Y shape and including a lift leg which has a lower end that is slidably connected to the base **302**. In such an example, the head end and foot end lift assemblies are selectively collapsible to adjust the height of the upper frame **304**.

[0133] The mattress **100** will now be described with reference to FIGS. 1 and 4 to 6A. As shown in FIG. 1, the mattress **100** extends longitudinally from a head end **102** to a foot end **105** and extends laterally from a left end **106** to a right end **108**. A length of the mattress **100** is defined between the head end **102** and the foot end **105**, while a width of the mattress **100** is defined between the left and right ends **106**, **108**. The mattress **100** has a cover **110** that defines the exterior surfaces of the mattress **100**. Notably, the cover **110** has a top surface **112** on which the patient is supported, and a bottom surface **114** (FIG. 6A) opposite the top surface **112**. A thickness of the mattress **100** is defined between the top and bottom surfaces **112**, **114**. In this embodiment, the top surface **112** is permeable to moisture and impermeable to air. In other embodiments, the top surface **112** may be permeable to air and impermeable to moisture. In use, the bottom surface **114** is in contact with the deck **306** of the bed **300**. In this embodiment, the cover **110** has a top portion **116** and a bottom portion **118** that are held together via a fastening system to contain the internal components of the mattress **100** that are disposed between the top and bottom surfaces **112**, **114**. In this example, the fastening system is a slide fastener (i.e., a zipper) that holds at least part of the top and bottom portions **116**, **118** of the cover **110** together. Other types of fastening systems are also contemplated.

[0134] Some of the internal components of the mattress **100** are shown in greater detail in FIGS. 4

to **6B**. As can be seen, the mattress **100** has a multitude of air-powered sections, including a plurality of inflatable sections that are selectively inflated and deflated to varying degrees in accordance with a desired configuration and/or function of the mattress **100**. In particular, with reference to FIG. **4**, the mattress **100** has an inflatable head section **120** for supporting the patient's head, an inflatable back section **122** for supporting the patient's torso, an inflatable seat section **123** for supporting the patient's upper legs, and an inflatable foot section **124** for supporting the patient's calves and feet. In this embodiment, the inflatable foot section **124** partly overlaps the inflatable seat section **123**. Each of the inflatable sections **120**, **122**, **123**, **124** includes a plurality of inflatable bladders **126** that can expand and contract in accordance with the degree of inflation thereof.

[0135] With reference to FIG. **11** which illustrates an example of a pneumatic diagram of the mattress **100**, the inflatable back section **122** includes a first inflatable back section **122A** and a second inflatable back section **122B** which can be pressurized separately to provide an alternating pressure therapy function. In particular, the inflatable bladders **126** of the inflatable back sections **122A**, **122B** are arranged alternately consecutively to each other such that each inflatable bladder **126** of the first section **122A** is disposed next to an inflatable bladder **126** of the second section **122B**. Similarly, the inflatable seat section **123** is divided into a first inflatable seat section **123A** and a second inflatable seat section **123B** which can be pressurized separately to provide an alternating pressure therapy function.

[0136] The inflatable bladders **126** can be made in any suitable way, including by welding two sheets (e.g., an upper sheet and a lower sheet) of material along predetermined locations to form the bladders **126** and communication channels between the bladders **126** such that bladders **126** that are part of a common pneumatic zone (i.e., a given inflatable section) are fluidly connected to each other via the communication channels. In this embodiment, the inflatable bladders **126** are elongated and generally extend either longitudinally or laterally. For instance, the inflatable bladders **126** of the inflatable back section **122**, inflatable seat section **123** and the inflatable foot section **124** extend laterally while the inflatable bladders **126** of the inflatable head section **120** extend longitudinally. The bladders **126** of the sections **120**, **122**, **123**, **124** may be oriented differently in other embodiments.

[0137] For each given inflatable section **120**, **122A**, **122B**, **123A**, **123B**, **124**, the inflatable bladders **126** thereof are fluidly connected to one another so that the inflatable bladders **126** of the given inflatable section are inflated and deflated together. As such, the inflatable sections **120**, **122A**, **122B**, **123A**, **123B**, **124** constitute separate pneumatic control zones.

[0138] Together, the inflatable sections **120**, **122**, **123**, **124** form a comfort layer of the mattress **100**. The pressure of the inflatable bladders **126** of the comfort layer is generally regulated to provide comfort to the patient as the inflatable bladders **126** of the comfort layer are the support components that are closest to the patient when the patient lies on the mattress **100**.

[0139] In this embodiment, as shown schematically in FIG. **11**, the mattress **100** also includes a first foot extension section **148** and a second foot extension section **149** that are selectively inflated to extend a length of the mattress **100**. Notably, when the foot extension sections **148**, **149** are both inflated, the length of the mattress **100** is at its maximum (e.g., 88 inches), and when the foot extension sections **148**, **149** are both deflated, the length of the mattress **100** is at its minimum (e.g., 80 inches). The foot extension sections **148**, **149** are separate pneumatic control zones such that it is also possible to only inflate the first foot extension section **148** and keep the second foot extension section **149** deflated, in which case the mattress **100** has an intermediate length (e.g., 84 inches) between the maximum and minimum lengths.

[0140] The mattress **100** also has a support layer disposed underneath the comfort layer to provide more rigidity to the mattress **100**. The comfort layer is generally softer than the support layer. With reference to FIG. **6A**, the support layer includes a seat support section **128** disposed underneath the inflatable seat section **123**. In this embodiment, the seat support section **128** is also inflatable and includes a plurality of support bladders **130** that are inflated and deflated according to a desired

rigidity of the mattress **100**. That is, the pressure within the support bladders **130** is regulated to provide adequate rigidity to the mattress **100** as may be necessary. In this embodiment, the support bladders **130** are elongated and extend longitudinally. The support bladders **130** are in fluid communication with each other such that they form a common pneumatic control zone.

[0141] With reference to FIG. **11**, the mattress **100** also has a back support section **136** and a foot support section **144** which are disposed underneath the inflatable back section **122** and the inflatable foot section **124** respectively. The back and foot support sections **136**, **144** are configured similarly to the seat support section **128** and have similar functions to the seat support section **128** described above but with respect to the inflatable back section **122** and the inflatable foot section **124** respectively.

[0142] It is contemplated that the support sections of the mattress **100** could be configured differently in other embodiments. For instance, in some embodiments, the support sections could instead include one or more foam members rather than including inflatable bladders.

[0143] With reference to FIGS. **4**, **6A** and **6B**, in this embodiment, left and right turning bladders **138** (schematically illustrated in FIG. **4**) are provided to turn a patient toward either lateral side of the mattress **100**. More specifically, the left and right turning bladders **138** are selectively inflated to turn the patient toward the right and left sides of the mattress **100** respectively. As shown in FIGS. **6A** and **6B**, in this embodiment, the left and right turning bladders **138** are disposed in part between the seat support section **128** and the inflatable seat section **123**. In other words, the turning bladders **138** are disposed above the seat support section **128** and below the inflatable seat section **123**. The turning bladders **138** also extend between the back support section **136** and the inflatable back section **122**. The left and right turning bladders **138** are disposed on opposite sides of a longitudinal centerplane of the mattress **100** (i.e., a vertical plane bisecting the width of the mattress **100**) and, as shown in FIG. **4**, each of the left and right turning bladders **138** extends along a torso portion of the mattress **100** (i.e., a portion of the mattress **100** that will generally be aligned with the torso of the patient) as well as part of a seat portion of the mattress **100**.

[0144] As shown in FIGS. **6A** and **6B**, the mattress **100** also has a structural foam layer **129** disposed underneath the seat support section **128** and the back support section **136**. The structural foam layer **129** could be omitted in some embodiments.

[0145] With continued reference to FIG. **6A**, in this embodiment, the mattress **100** also has outer lateral support bladders **132** disposed laterally outwardly from the support bladders **130**. Each of the outer lateral support bladders **132** sits underneath an outer lateral portion of the inflatable back section **122**. The outer lateral support bladders **132** are disposed atop a respective foam support block **133** that extends longitudinally. The foam support blocks **133** may house some components such as pneumatic tubing for routing air to and from some of the inflatable sections of the mattress **100**. The foam support blocks **133** may also provide additional rigidity to the mattress **100**. As can be seen, when inflated at their operating pressure, the outer lateral support bladders **132** have a cross-sectional profile having a height that is smaller than a height of the support bladders **130** when the latter are at their operating pressure.

[0146] It is contemplated that the outer lateral support bladders **132** could be omitted in other embodiments. For instance, in some embodiments, the outer lateral support bladders **132** could instead be replaced by additional support bladders **130**, and the foam support blocks **133** could be omitted.

[0147] As shown in FIGS. **4** and **5**, in this embodiment, to each side of the inflatable back section **122** and the inflatable seat section **123**, there is disposed a respective inflatable lateral extension **140** that spans a majority of the length of the mattress **100**. The lateral extensions **140** are selectively inflated to adjust a width of the mattress **100**. Notably, the lateral extensions **140** can be deflated such that respective bladders of the lateral extensions **140** collapse (as shown in FIG. **6B**), thereby reducing the width of the mattress **100**, and conversely the lateral extensions **140** can be inflated (as shown in FIGS. **4** to **6A**) to increase the width of the mattress **100** to approximately a

45-inch width. This may be useful for example to accommodate bariatric patients. Moreover, this feature may be used in conjunction with a width extending function of the bed **300**. In some embodiments, the lateral extensions **140** may be omitted (i.e., the width of the mattress **100** may not be adjustable). Each lateral extension **140** has a plurality of lateral extension bladders **142** that are disposed side-by-side in the longitudinal direction of the mattress **100**. For each lateral extension **140**, the lateral extension bladders **142** thereof are in fluid communication with each other via communication channels **143** defined between consecutive ones of the lateral extension bladders **142**.

[0148] In some embodiments, the lateral extension **140** may be a first lateral extension, and a second lateral extension may be provided laterally outwardly of the first lateral extension. In such cases, the first lateral extension may be inflated to achieve a first extended width of the mattress **100**, and the second lateral extension may be inflated to achieve a second extended width of the mattress **100** greater than the first extended width.

[0149] As shown in FIG. 6A, the mattress **100** also has inflatable lateral support sections **134** disposed underneath respective ones of the lateral extensions **140**. Each lateral support section **134** includes a plurality of lateral support bladders **135** (only one of which is shown in FIG. 6A) that are disposed side-by-side in the longitudinal direction of the mattress **100**. The lateral support sections **134** provide additional rigidity to the mattress **100** at the lateral outer portions thereof when the lateral extensions **140** are deployed. To that end, the lateral support bladders **135** are selectively inflatable together with the lateral extension bladders **142** of the lateral extensions **140**. In other words, the lateral support bladders **135** are inflated (as shown in FIG. 6A) together with the lateral extension bladders **142**, and are deflated (as shown in FIG. 6B) together with the lateral extension bladders **142**. When the lateral extensions **140** and the lateral support sections **134** are deployed, a pressure to which the lateral support bladders **135** are pressurized is greater than the pressure within the lateral extension bladders **142** in order to provide a rigid support below the lateral extension bladders **142**. In some cases, the pressure in the lateral extension bladders **142** may be adjusted, namely reduced, to decrease a thickness of the mattress **100** at the lateral outer side thereof and thereby facilitate a patient exiting the bed from the side, while a pressure within the lateral support bladders **135** remains constant to provide a rigid support underneath the lateral extension bladders **142**. It is contemplated that, in some embodiments, the lateral support bladders **135** could be omitted (e.g., the lateral extension bladders **142** could instead extend vertically to the level of the support bladders **130**).

[0150] In this embodiment, the left and right lateral extensions **140** are in fluid communication with each other via a fluid connection (e.g., pneumatic tubing). As such, in this example, the left and right lateral extensions **140** are deployed and collapsed simultaneously. In other words, the left and right lateral extensions **140** form a common pneumatic zone. It is contemplated that, in other embodiments, the left and right lateral extensions **140** could be fluidly independent from each other (i.e., they could be independent pneumatic zones) such that the left and right lateral extensions **140** could be deployed or collapsed independently from each other.

[0151] In this embodiment, the mattress **100** also has a dorsal push bladder **145** (shown schematically in FIG. 11) configured to be selectively inflated to help the patient out of the mattress **100**. More specifically, the dorsal push bladder **145** is inflated in response to the bed **300** moving to its chair egress position (FIG. 3) in order to assist the patient to move forward when they are exiting the bed **300** from the chair egress position. The dorsal push bladder **145** may be disposed underneath the back support section **136** and extends along a torso portion of the mattress **100** (i.e., the part of the mattress **100** intended to be aligned with a torso of the patient). The dorsal push bladder **145** could be omitted in other embodiments.

[0152] Furthermore, as illustrated in hidden lines in FIG. 5, in this embodiment, the mattress **100** also has a microclimate management layer **200** configured to diffuse air towards the top surface **112** of the mattress **100**. This may help improve patient comfort, namely by keeping the patient

cool and dry or by eliminating humidity that has penetrated the mattress **100**. The microclimate management layer **200** is disposed above the inflatable back section **122**. In other words, the microclimate management layer **200** is disposed between the inflatable back section **122** and the top surface **112**. The microclimate management layer **200** defines in part a pocket configured to receive air therein. The pocket may define openings **165** through which air flows above the comfort layer of the mattress **100**. The microclimate management layer **200** thus provides a low air loss function of the mattress **100**. The microclimate management layer **200** may be omitted in other embodiments.

[0153] A pneumatic control assembly **150**, illustrated in hidden lines in FIG. 5, is provided to distribute air to the various air-powered parts of the mattress **100**, including the various inflatable sections and the microclimate management layer **200** described above. The pneumatic control assembly **150** is disposed along a foot portion of the mattress **100**, namely underneath the foot support section **144** and the inflatable foot section **124**. It is contemplated that, in other embodiments, the pneumatic control assembly **150** could be located elsewhere along the mattress **100** (e.g., along the head portion of the mattress **100**).

[0154] As shown in FIG. 7, the pneumatic control assembly **150** has an external housing **152** for enclosing the internal components of the pneumatic control assembly **150**. The housing **152** has a base portion **154** and a cover portion **156** that is removably connected to the base portion **154**. The housing **152** has a foot end **158** and a head end **160** which are longitudinally spaced from each other, and a left end **162** and a right end **164** which are laterally spaced from each other. The housing **152** has a larger width, defined between left and right ends **162**, **164**, than a length, defined between foot and head ends **158**, **160**. Bumpers (not shown) may be provided at the left and right ends **162**, **164** of the housing **152** to protect the pneumatic control assembly **150** from impacts. As will be described in more detail below, the housing **152** defines internal channels that communicate different parts of the pneumatic control assembly **150** with each other.

[0155] The pneumatic control assembly **150** has an air supply device that pressurizes air for distribution to the different air-powered parts of the mattress **100**. More particularly, as shown in FIGS. 8 and 11, in this embodiment, the pneumatic control assembly **150** has a plurality of air supply devices including two blowers **166.sub.1**, **166.sub.2** that are in selective communication with different parts of the mattress **100** to feed air thereto or aspirate the air contained thereby. Blowers provide a high flow rate of air which can be useful for rapidly inflating the different bladders of the mattress **100**, thereby reducing the time necessary for the mattress **100** to reach adequate pressure levels (e.g., at initialization of the mattress **100** or to accommodate a different position of the bed **300**). For instance, the blowers **166.sub.1**, **166.sub.2** have a flow rate that is greater than 100 Lpm. In this example, the blowers **166.sub.1**, **166.sub.2** have a flow rate of approximately 500 Lpm. However, while blowers can supply large volumes of air quickly, their capacity to pressurize the air is relatively limited. For instance, the blowers **166.sub.1**, **166.sub.2** may have a pressure capacity that is less than 15 kPa. In other words, the blowers **166.sub.1** may pressurize the air between 0 and 15 kPa. More specifically, in this example, the blowers **166.sub.1**, **166.sub.2** each have a pressure capacity of approximately 5.4 kPa (i.e., approximately 42 mm Hg). The blowers **166.sub.1**, **166.sub.2** thus supply air at a high flow rate and at a low pressure. As will be described in greater detail below, in this embodiment, the pneumatic control assembly **150** can selectively operate in a dual-stage compression mode whereby air is made to be pressurized consecutively by each blower **166.sub.1**, **166.sub.2** before distributing air to one or more of the air-powered parts of the mattress **100**. This may allow the air to reach a greater pressure than would be possible if a single blower were to supply that air.

[0156] As shown in FIG. 7, in this embodiment, each blower **166.sub.1**, **166.sub.2** is enclosed within a respective blower enclosure **186** that, together with the base portion **154** of the housing **152**, defines a blower chamber **187** (FIG. 8). In use, air is drawn into each blower chamber **187** for distribution by the enclosed blower to different parts of the mattress **100**. With reference to FIGS.

29 and **30** which illustrate an exemplary air supply unit including a given one of the blowers **166.sub.1** enclosed within a corresponding blower enclosure **186**, the blower enclosure **186** has two housing inlets **402**, **404** through which air enters the blower enclosure **186** and a housing outlet **406** through which air exits the blower enclosure **186**. Elbow connectors **409** may be connected to the housing inlets and outlets **402**, **404**, **406** for connection to other parts of the pneumatic control assembly **150** as will be explained in greater detail further below. A peripheral flange **408** of the blower enclosure **186** is affixed to the base portion **154** of the housing **152** by fasteners **410**. A gasket **412** is disposed at an interface between the blower enclosure **186** and the base portion **154** to provide a seal therebetween.

[0157] The air supply unit may be configured to limit the noise emitted by the operation of the corresponding blower **166.sub.1**, **166.sub.2**. More specifically, with particular reference to FIGS. **31** and **32**, in this embodiment, the air supply unit includes an attenuating filler **414** that surrounds the blower **166.sub.i** to fix the blower **166.sub.1** in place within the blower chamber **187**. The blower **166.sub.i** is therefore not retained in place by fasteners as is typical in conventional pneumatic systems. For example, the blower **166.sub.i** is detached from the base portion **154** of the housing **152**. The blower **166.sub.i** is rather held in place by the attenuating filler **414** filling a substantial amount of space within the blower chamber **187** so that the blower **166.sub.i** cannot move due to the presence of the attenuating filler **414** extending from an outer surface of the blower **166.sub.i** to an inner surface of the blower enclosure **186**. In this embodiment, the attenuating filler **414** is made of a plurality of filler members including an upper filler member **416** and a lower filler member **418** which are disposed respectively above and below the blower **166.sub.i** to complementarily surround the blower **166.sub.i**. As shown in FIGS. **31** and **32**, the filler members **416**, **418** define respective recesses **420**, **422** corresponding in part to a shape of the blower **166.sub.i** so that the outer surfaces of the blower **166.sub.i** partly match the inner surfaces of the filler members **416**, **418** that define the recesses **420**, **422**. To that end, the recesses **420**, **422** may be produced by molding the filler members **416**, **418** to be a negative of the shape of the blower **166.sub.i**. The filler members **416**, **418** are also shaped to allow communication between the housing inlets **402**, **404** and an inlet **430** (FIG. **31**) of the blower **166.sub.i**, and between the housing outlet **406** and an outlet **432** (FIGS. **31**, **32**) of the blower **166.sub.i**. Notably, the filler members **416**, **418** may form, together, channels through which air can freely flow to and from the blower inlet **430** and outlet **432**. The attenuating filler **414** is made of a soft material that can attenuate vibration. In this embodiment, the attenuating filler **414** is made of foam. The foam of the attenuating filler **414** may have a density of approximately 7 lbs/ft.^{sup.3}. In this example, the foam of the attenuating filler **414** has flame retardant properties. The attenuating filler **414** minimizes the vibrations of the blower **166.sub.i** that could otherwise generate considerable noise if not attenuated.

[0158] While two filler members are provided in this embodiment, it is contemplated that a single filler member could be provided in other embodiments. For example, in some cases, a single filler member may be provided to overlie the blower **166.sub.i**. In such cases, the blower **166.sub.i** may be secured in place by a complementary shape of the filler member and the base portion **154** of the housing **152**. Furthermore, a similar concept may be implemented for air supply devices other than a blower.

[0159] Furthermore, as shown in FIG. **32**, in this embodiment, a flexible sleeve **434** connects the outlet **432** of the blower **166.sub.i** to the housing outlet **406**. The flexible sleeve **434** is made of a flexible fabric (e.g., spandex coated with thermoplastic polyurethane (TPU)). The flexible sleeve **434** could be fixed in place in various ways. In this example, one end of the flexible sleeve **434** is welded to the outlet **432** of the blower **166.sub.i** to surround the outlet **432**, while another end of the flexible sleeve **434** is welded to the blower enclosure **186** at the housing outlet **406**. This provides an airtight connection between the blower **166.sub.i** and the housing outlet **406**. The flexible sleeve **434** minimizes vibrations flow that might otherwise be generated with a more rigid

connector.

[0160] It is contemplated that, in other embodiments, other features may be disposed within the blower chamber **187**, such as baffles for example, to limit the noise and/or vibration generated by the operation of the blowers **166.sub.1**, **166.sub.2**.

[0161] With reference to FIGS. **9** and **11**, the pneumatic control assembly **150** has another air supply device, namely a compressor **168**. The compressor **168** is used for piloting valves that selectively communicate the blowers **166.sub.1**, **166.sub.2** with the different air-powered parts of the mattress **100**. Compressors, in contrast with blowers, provide a low flow rate of air but at a high pressure. For instance, the compressor **168** may have a pressure capacity that is greater than 50 kPa. The pressure capacity of the compressor **168** may thus be 10 times or more greater than the pressure capacity of the blowers **166.sub.1**, **166.sub.2**. In particular, in this embodiment, the compressor **168** has a pressure capacity of approximately 100 kPa (i.e., approximately 1 bar). The compressor **168** may have a different pressure capacity in other embodiments. For instance, the pressure capacity of the compressor **168** may be between 50 kPa and 100 kPa inclusively. Furthermore, the flow rate of the compressor **168** may be less than 10 Lpm. In this example, the flow rate of the compressor **168** is approximately 5 Lpm. The flow rate of the compressor **168** may thus be 100 times less than the flow rate of the blowers **166.sub.1**, **166.sub.2**. The compressor **168** therefore supplies air at a greater pressure than the blowers **166.sub.1**, **166.sub.2** but at a lower flow rate than the blowers **166.sub.1**, **166.sub.2**. As will be described in greater detail below, in this embodiment, the compressor **168** may also be selectively used to, under certain conditions, feed air to the different air-powered parts of the mattress **100** (instead of being exclusively used for piloting valves).

[0162] With reference to FIG. **9**, in this embodiment, the compressor **168** is disposed within a compressor chamber **169** that is defined by the base portion **154** of the housing **152** on an underside thereof. A lid (not shown) is connected to the base portion **154** of the housing **152** to close off the compressor chamber **169**. In use, when the compressor **168** is activated, air from an exterior of the pneumatic control assembly **150** is drawn into the compressor chamber **169** via an inlet aperture (not shown). A muffler may also be disposed within the compressor chamber **169** to limit the noise emitted by operation of the compressor **168**.

[0163] The compressor **168** may be electrically controlled in such a manner as to limit noise emitted thereby. Notably, in this embodiment, as exemplified by the graph of FIG. **23**, the compressor **168** is operated at an applied voltage AV that is less than a rated input voltage RIV of the compressor **168**. For example, the rated input voltage RIV may be approximately 24V. When starting the compressor **168**, the applied voltage AV is increased to a peak voltage that may be equal to or less than the rated input voltage RIV to overcome an inertia of the compressor **168**. The applied voltage AV is then lowered to a nominal voltage NV that is determined based on the output pressure required from the compressor **168**. The nominal voltage NV may be significantly less than the rated input voltage RIV. For instance, according to one example, the nominal voltage NV may be 12V. By lowering the applied voltage AV to a value that is less than the rated input voltage RIV, the noise emitted by the compressor **168** may be limited. In this embodiment, the applied voltage AV of the compressor **168** is pulse width modulated.

[0164] The compressor **168** may be controlled conventionally in other embodiments (i.e., the applied voltage AV is either null or the rated input voltage RIV).

[0165] In this embodiment, the pneumatic control assembly **150** also includes reservoirs of pressurized air that are actively filled by one of the air supply devices. For instance, as shown in FIG. **11**, the pneumatic system of the pneumatic control assembly **150** may include reservoirs **230**, **231**, **233** that are selectively filled by the compressor **168**. The reservoirs **230**, **231**, **233** are provided to help maintain a more constant pressure in the pneumatic circuit that is fed by the compressor **168** (which may be referred to as a “high-pressure circuit”). In addition, the reservoirs **230**, **231**, **233** may also be provided as a fail-safe to close a “low-pressure circuit” of the pneumatic

system (i.e., the circuit associated with the blowers **166.sub.1**, **166.sub.2**) in the event of a malfunction of the compressor **168** and/or of a valve. This may be helpful to ensure a minimum inflation level for at least some of the inflatable bladders of the mattress **100** in order to prevent the patient coming into contact with a rigid component of the bed **300**.

[0166] As shown in FIGS. **7** and **11**, in order to direct air from the air supply devices to the different air-powered parts of the mattress **100** (and vice-versa), the pneumatic control assembly **150** has first and second manifolds **170.sub.1**, **170.sub.2** that are associated with the first and second blowers **166.sub.1**, **166.sub.2** respectively. The manifolds **170.sub.1**, **170.sub.2** are provided for fluidly connecting the air supply devices to the various air-powered parts of the mattress **100**. Each manifold **170.sub.1**, **170.sub.2** includes a plurality of manifold valves **172.sub.1-172.sub.9** that control air flow in and out of the air-powered parts of the mattress **100**. Notably, each of the manifold valves **172.sub.1-172.sub.9** is associated with a corresponding air-powered part of the mattress **100** and is selectively fluidly connected thereto. By selectively opening the manifold valves **172.sub.1-172.sub.9**, the associated air-powered parts of the mattress **100** are placed in fluid communication with a corresponding air supply device. As can be seen, in some cases, some inflatable bladders of the mattress **100** may be associated with more than one manifold valve. For instance, in this example, each of the left and right turning bladders **138** is selectively fluidly connected to the air supply devices by two of the manifold valves **172.sub.1-172.sub.9**. This may allow a significant flow rate of air to be directed to the turning bladders **138** by opening both manifold valves that are associated therewith simultaneously. Alternatively, a single manifold valve could be associated therewith and have a configuration that allows a greater flow rate of air (e.g., a bigger outlet opening).

[0167] The manifolds **170.sub.1**, **170.sub.2** also include respective inlet valves **176.sub.1**, **176.sub.2**, interconnection valves **178.sub.1**, **178.sub.2** and control valves **179.sub.1**, **179.sub.2**. The inlet valves **176.sub.1**, **176.sub.2** selectively fluidly connect the blowers **166.sub.1**, **166.sub.2** to respective inlet apertures (not shown) through which exterior air is drawn into the pneumatic control assembly **150**. As will be discussed below, the inlet valves **176.sub.1**, **176.sub.2** may additionally be used to exhaust air therethrough to the exterior of the pneumatic control assembly **150** via the respective inlet apertures. The interconnection valves **178.sub.1**, **178.sub.2** are provided to selectively fluidly interconnect the manifolds **170.sub.1**, **170.sub.2** to each other. The control valves **179.sub.1**, **179.sub.2** are provided for selectively blocking the inlets of the respective blowers **166.sub.1**, **166.sub.2**. This may be useful in certain operations to exhaust air for example via the inlet valves **176.sub.1**, **176.sub.2** as will be described in more detail below. In this embodiment, as shown in FIG. **11**, the pneumatic control assembly **150** includes solenoid valves **201.sub.1**, **201.sub.2**, **203.sub.1**, **203.sub.2** for controlling air flow to respective pilot ports of the valves **176.sub.1**, **176.sub.2**, **178.sub.1**, **178.sub.2**. Other solenoid valves (not shown) control air flow from the compressor **168** and/or reservoir **230** to the pilot ports of the control valves **179.sub.1**, **179.sub.2**. More specifically, the inlet valves **176.sub.1**, **176.sub.2** are pneumatically actuated by the compressor **168** or reservoir **230** via the solenoid valves **201.sub.1**, **201.sub.2** that are controlled by controller **180** (FIG. **8**). For example, when the solenoid valve **201.sub.1** is closed, air flow from the compressor **168** fed into a pilot port of the first inlet valve **176.sub.1** ceases, causing the first inlet valve **176.sub.1** to open. Similar operations are executed to open the other inlet valve **176.sub.2** via the solenoid valve **201.sub.2**, the interconnection valves **178.sub.1**, **178.sub.2** via the solenoid valves **203.sub.1**, **203.sub.2**, and control valves **179.sub.1**, **179.sub.2** via other solenoid valves (not shown) that control air flow to the pilot ports of the control valves **179.sub.1**, **179.sub.2**. The solenoid valves controlling air flow to the pilot ports of the control valves **179.sub.1**, **179.sub.2** may be grouped with the other solenoid valves **201.sub.1**, **201.sub.2**, **203.sub.1**, **203.sub.2**, **205**, **207** in the pneumatic diagram of FIG. **11**. Some of the above-mentioned valves may be omitted in some embodiments. For instance, it is contemplated that the control valves **179.sub.1**, **179.sub.2** may be omitted in some embodiments.

[0168] The construction of the manifolds **170.sub.1**, **170.sub.2** will be described in greater detail further below.

[0169] With reference to FIGS. **7** and **8**, the controller **180** of the pneumatic control assembly **150** is in communication with the blowers **166.sub.1**, **166.sub.2** and the compressor **168** to control their operation. The controller **180** is configured to receive different inputs and, in response to those inputs, control the air supply devices. The controller **180** has a processor unit **182** for carrying out executable code, and a non-transitory memory unit **184** that stores the executable code in a non-transitory medium (not shown) included in the memory unit **184**. The processor unit **182** includes one or more processors for performing processing operations that implement functionality of the controller **180**. The processor unit **182** may be a general-purpose processor or may be a specific-purpose processor comprising one or more preprogrammed hardware or firmware elements (e.g., application-specific integrated circuits (ASICs), electrically erasable programmable read-only memories (EEPROMs), etc.) or other related elements. The non-transitory medium of the memory unit **184** may be a semiconductor memory (e.g., read-only memory (ROM) and/or random-access memory (RAM)), a magnetic storage medium, an optical storage medium, and/or any other suitable type of memory. While the controller **180** is represented as being one entity in this implementation, it is understood that the controller **180** could comprise separate entities for controlling components separately.

[0170] The controller **180** is also in communication with and controls a plurality of solenoid valves **174** of the pneumatic control assembly **150**. Each solenoid valve **174** is associated with a corresponding manifold valve **172.sub.1**. As shown in FIGS. **7** and **8**, in this embodiment, there are two subsets **175.sub.1**, **175.sub.2** of the solenoid valves **174**, the solenoid valves **174** of each subset **175.sub.1**, **175.sub.2** being in communication with the manifold valves **172.sub.1-172.sub.9** of a respective manifold **170.sub.1**, **170.sub.2**. Notably, each solenoid valve **174** is connected between a pilot port of a respective manifold valve **172.sub.i** and the compressor **168**. The solenoid valves **174** are actuated by the controller **180** to selectively allow air flow from the compressor **168** to the pilot ports of the manifold valves **172.sub.1-172.sub.9**, thus causing the compressor **168** to actuate the manifold valves **172.sub.i** as determined by the controller **180**. Each solenoid valve **174** is fluidly connected to a corresponding manifold valve **172.sub.i** via tubing (not shown in FIGS. **7** and **8**) extending between an outlet of the solenoid valve **174** and the pilot port of a manifold valve **172.sub.i**. The solenoid valves **174** are normally open and therefore, without intervention from the controller **180** to actuate the solenoid valves **174**, air flow from the compressor **168** (or reservoir **230**) enters the respective pilot ports of the manifold valves **172.sub.1-172.sub.9**, thereby closing the manifold valves **172.sub.1-172.sub.9**. However, upon actuation of a given solenoid valve **174** to close that solenoid valve **174**, the air flow to the pilot port of the corresponding manifold valve **172.sub.i** is cut off and consequently the manifold valve **172.sub.i** opens thereby fluidly communicating the corresponding blower **166.sub.i** with the inflatable section of the mattress **100** associated with that manifold valve **172.sub.i**. For example, if the controller **180** has determined that the pressure of the inflatable seat section **123B** is to be increased, the controller **180** actuates the solenoid valve **174** that is in selective fluid communication with the manifold valve **172.sub.1** in order for the air flow from the compressor **168** to the pilot port of the manifold valve **172.sub.1** to be stopped, thereby opening the manifold valve **172.sub.1** so that air from the blower **166.sub.1** can flow through the manifold valve **172.sub.1** to the inflatable seat section **123B**.

[0171] The controller **180** is also in communication with the solenoid valves **201.sub.1**, **201.sub.2**, **203.sub.1**, **203.sub.2** as well as with the solenoid valves that control air flow to the pilot ports of control valves **179.sub.1**, **179.sub.2**. Furthermore, with reference to FIG. **11**, the reservoir **233** is provided to ensure that the solenoid valves **201.sub.1**, **201.sub.2**, **203.sub.1**, **203.sub.2**, **205**, **207** (and the solenoid valves controlling air flow to the control valves **179.sub.1**, **179.sub.2**) are provided with a sufficient quantity of air to close the valves **176.sub.1**, **176.sub.2**, **178.sub.1**, **178.sub.2**, **179.sub.1**, **179.sub.2**, **220**, **222** in case of failure of one or more of the solenoid valves

174 for example. This would close off fluid communication of the pneumatic control assembly **150** with the exterior via the inlet valves **176.sub.1**, **176.sub.2** and the valves **220**, **222** and thereby ensure that the air is not emptied from mattress **100**. This may be useful to prevent the patient from coming into contact with a rigid component of the bed **300** in case of a component failure. Moreover, a check valve **177** collaborates with the reservoir **233** to ensure that the volume of air of the reservoir **233** remains available to the valves **201.sub.1**, **201.sub.2**, **203.sub.1**, **203.sub.2**, **205**, **207**. Similarly, the reservoir **231** is provided to ensure that the solenoid valves **174** are provided with a sufficient quantity of air to close the manifold valves **172.sub.1-172.sub.9** in case of failure of one or more of the solenoid valves **201.sub.1**, **201.sub.2**, **203.sub.1**, **203.sub.2**, **205**, **207** (and the solenoid valves controlling air flow to the control valves **179.sub.1**, **179.sub.2**). Moreover, a check valve **181** collaborates with the reservoir **231** to ensure that the volume of air of the reservoir **231** remains available to the solenoid valves **174**. In this embodiment, the reservoirs **231**, **233** have a smaller volume than the reservoir **230**.

[0172] The pneumatic control assembly **150** is operable in multiple operation modes that are selected by the controller **180**. Notably, the controller **180** determines which operation mode to engage depending on the desired outcome at one or more air-powered parts of the mattress **100**. In particular, in this embodiment, the operation modes that the controller **180** selectively implements include a direct feed mode, a dual-stage compression mode, a compressor feed mode and a deflate mode. One or more of these operation modes may be omitted in some embodiments.

[0173] FIG. **10** illustrates an example of a method **1000** executed by the controller **180** to choose which of the operation modes to implement in order to achieve a desired target pressure $P_{sub.x}$ at the inflatable bladders of a given inflatable section of the mattress **100**. The target pressure $P_{sub.x}$ may be any suitable pressure that the controller **180** has determined, at step **1010**, should be implemented at a given inflatable section (e.g., the bladders of the back support section **136**). The target pressure $P_{sub.x}$ may be determined based on one or more inputs such as a command received from a user input device such as an “increase firmness” button displayed on a touch screen of the bed or a “chair position” button of the bed for causing the bed to assume the chair position. In other cases, the target pressure $P_{sub.x}$ may be a pre-determined target pressure that is programmed into the controller **180** such as a start-up pressure that is to be implemented in the given inflatable section at initialization of the mattress **100**. The skilled reader will understand that the target pressure $P_{sub.x}$ may be determined based on different variables in other embodiments.

[0174] With the target pressure $P_{sub.x}$ determined, the method proceeds to step **1020** where the controller **180** compares the target pressure $P_{sub.x}$ to a current pressure $P_{sub.c}$ in the given inflatable section to determine if the target pressure is equal to the current pressure $P_{sub.c}$. The current pressure $P_{sub.c}$ of the given inflatable section may be detected by a pressure sensor (e.g., sensors **225** in FIG. **11**) associated with the given inflatable section and communicated to the controller **180**. If the target pressure $P_{sub.x}$ is equal to the current pressure $P_{sub.c}$ (i.e., no pressure adjustment needed), the method returns to step **1010** to proceed with the next pressure adjustment of an inflatable section of the mattress **100**. The target pressure $P_{sub.x}$ may have a defined tolerance window (e.g., ± 2 mm Hg) such that, if the current pressure $P_{sub.c}$ is within the tolerance window of the target pressure $P_{sub.x}$, the current pressure $P_{sub.c}$ is determined to be equal to the target pressure $P_{sub.x}$.

[0175] If the target pressure $P_{sub.x}$ is determined to be different from the current pressure $P_{sub.c}$, the method proceeds to step **1030**. At step **1030**, the controller **180** determines if the target pressure $P_{sub.x}$ is greater than the current pressure $P_{sub.c}$. If it is not (i.e., the target pressure $P_{sub.x}$ is less than the current pressure $P_{sub.c}$), the method proceeds to step **1040** and the deflate mode is implemented by the controller **180** to decrease the pressure within the given inflatable section.

[0176] In the deflate mode, one of the blowers **166.sub.1**, **166.sub.2** draws air from the given inflatable section and drives it outside of the pneumatic system of the mattress **100**. As will be described in more detail below, in this embodiment, the air that is aspirated from an inflatable

section of the mattress **100** associated with the first manifold **170.sub.1** may be evacuated via the inlet valve **176.sub.1** of the first manifold **170.sub.1** and similarly, the air that is aspirated from an inflatable section of the mattress **100** associated with the second manifold **170.sub.2** may be evacuated via the inlet valve **176.sub.2** of the second manifold **170.sub.2**. The air drawn from the given inflatable section may be discharged via both the inlet valves **176.sub.1**, **176.sub.2** when a large volume of air is to be evacuated from the given inflatable section. In some cases, the air drawn from the given inflatable section may be driven to a purge valve **220** (FIG. **11**) that fluidly connects the blowers **166.sub.1**, **166.sub.2** to the exterior of the pneumatic control assembly **150**. In some cases, the air drawn from the given inflatable section may be driven to an internal space of the mattress **100** or a component of the mattress **100** other than an inflatable section via a discharge valve. For instance, the discharge valve may be a microclimate management valve **222** which is fluidly connected to the microclimate management layer **200** in order to drive the air thereto. This may be the case for example when a smaller volume of air is to be evacuated from the given inflatable section.

[0177] In this embodiment, the purge valve **220** and the microclimate management valve **222** are piloted by air flow from the compressor **168**, or from the reservoir **230** that is filled by the compressor **168**, via respective solenoid valves **207**, **205** that are in communication with the controller **180**. That is, the solenoid valves **205**, **207** are selectively actuated to allow or restrict air flow from the compressor **168** or reservoir **230** to pilot ports of the microclimate management valve **222** and the purge valve **220** which causes actuation thereof. The reservoir **230** contains pressurized air thus providing a passive air supply that can actuate the air piloted valves of the pneumatic system. This may reduce the number of times the compressor **168** is activated during operation and help maintain a more uniform pressure in the high-pressure circuit of the pneumatic system. The reservoir **230** is selectively filled by the compressor **168** by actuating a control valve **185** (FIG. **11**) controlled by controller **180** and which, when actuated, directs air flow from the outlet of the compressor **168** to the reservoir **230**. It is contemplated that the reservoir **230** may be omitted in some embodiments.

[0178] With continued reference to FIG. **11**, an example of operation of the pneumatic control assembly **150** in the deflate mode will now be described according to a scenario in which the second inflatable seat section **123B**, associated with the manifold valve **172.sub.1** of the first manifold **170.sub.1**, is to be depressurized. First, the manifold valve **172.sub.1** is opened to allow air to be drawn therethrough from the inflatable seat section **123B**. This is done by actuating the corresponding solenoid valve **174** of the subset **175.sub.1** to cease air flow from the compressor **168** or reservoir **230** to the pilot port of the manifold valve **172.sub.1**. Then, the second interconnection valve **178.sub.2** and the second control valve **179.sub.2** are opened to communicate the opened manifold valve **172.sub.1** of the first manifold **170.sub.1** with the second blower **166.sub.2**, and the second blower **166.sub.2** is activated. With the manifold valves **172.sub.1**-**172.sub.9** of the second manifold **170.sub.2** closed and the first blower **166.sub.1** deactivated, the first interconnection valve **178.sub.1** is opened to communicate the second blower **166.sub.2** with the first inlet valve **176.sub.1** which is opened, while the first control valve **179.sub.1** is closed. The air drawn from the inflatable seat section **123B** is therefore discharged out of the pneumatic control assembly **150** through the inlet aperture associated with the first inlet valve **176.sub.1**. It is to be understood that the order of operations in the sequence may vary in other embodiments. The depressurization of the inflatable seat section **123B** is stopped once a pressure sensor **225** associated with the inflatable seat section **123B** indicates that the pressure in the inflatable seat section **123B** has reached the target pressure $P_{sub.x}$.

[0179] As mentioned above, in some cases, the air drawn from the inflatable section may be discharged via the microclimate management valve **222** or the purge valve **220**. In such instance, instead of operating the valves as discussed above, the first interconnection valve **178.sub.1** and inlet valves **176.sub.1**, **176.sub.2** are closed, and the first blower **166.sub.1** deactivated. The second

interconnection valve **178.sub.2** and the second control valve **179.sub.2** are opened to allow air flow therethrough, thereby fluidly interconnecting the manifold **170.sub.1** with the inlet of the second blower **166.sub.2**. The second blower **166.sub.2** is then activated which causes the second blower **166.sub.2** to draw air from the inflatable seat section **123B**. One of the purge valve **220** and the microclimate management valve **222** is then opened to allow the second blower **166.sub.2** to drive the air drawn from the inflatable seat section **123B** thereto. It is to be understood that the order of operations in the sequence may vary in other embodiments.

[0180] The same process may be implemented to depressurize the other inflatable sections associated with the manifold valves **172.sub.2-172.sub.9** of the first manifold **170.sub.1**.

Conversely, if an inflatable section associated with the second manifold **170.sub.2** is to be depressurized in the deflate mode, the same process can be carried out by controlling counterpart components associated with the first and second blowers **166.sub.1**, **166.sub.2** in the same manner. For instance, if the inflatable foot section **124** associated with the manifold valve **1725** of the second manifold **170.sub.2** has to be depressurized, the manifold valve **1725** of the second manifold **170.sub.2** is opened, the first interconnection valve **178.sub.1** and the first control valve **179.sub.1** are opened to communicate the opened manifold valve **1725** of the second manifold **170.sub.2** with the first blower **166.sub.1**, and the first blower **166.sub.1** is activated. With the manifold valves **172.sub.1-172.sub.9** of the first manifold **170.sub.1** closed and the second blower **166.sub.2** deactivated, the second interconnection valve **178.sub.2** is opened to communicate the first blower **166.sub.1** with the second inlet valve **176.sub.2** which is opened, while the second control valve **179.sub.2** is closed. The air drawn from the inflatable foot section **124** is therefore discharged out of the pneumatic control assembly **150** through the inlet aperture associated with the second inlet valve **176.sub.2**. Alternatively, the air can be driven to the microclimate management valve **222** or the purge valve **220** in a similar manner to that describe above.

[0181] The purge valve **220** and the microclimate management valve **222** may be integrated as part of the manifolds **170.sub.1**, **170.sub.2** in other embodiments. For instance, each manifold **170.sub.1**, **170.sub.2** may have a dedicated purge valve **220**. In yet other embodiments, the purge valve **220** may be omitted. In such cases, large volumes of air may be discharged exclusively via the inlet valves **176.sub.1**, **176.sub.2** as described above.

[0182] Returning now to FIG. **10**, if at step **1030** the target pressure $P_{sub.x}$ is determined to be greater than the current pressure $P_{sub.c}$, the method proceeds to step **1050**. At step **1050**, the controller **180** compares the pressure adjustment needed to the current pressure $P_{sub.c}$ of the given inflatable section to a threshold pressure adjustment $\Delta P_{sub.lim}$. More specifically, the controller **180** determines if a difference between the target pressure $P_{sub.x}$ and the current pressure $P_{sub.c}$ is greater than the threshold pressure adjustment $\Delta P_{sub.lim}$. As such, the controller **180** takes into account the magnitude of the pressure adjustment that is needed for the current pressure $P_{sub.c}$ to reach the target pressure $P_{sub.x}$. If the pressure adjustment is equal to or less than the threshold pressure adjustment $\Delta P_{sub.lim}$ (i.e., if the pressure adjustment is small enough), the controller **180** proceeds to step **1060** where it implements the compressor feed mode whereby the air flow directed to the given inflatable section is provided by the compressor **168**. In this example, the threshold pressure adjustment is 3 mm Hg.

[0183] In the compressor feed mode, the compressor **168** drives air to the manifolds **170.sub.1**, **170.sub.2**. To that end, in the compressor feed mode, the control valve **185** is in a position to direct air flow from the outlet of the compressor **168** to two fluid lines **187.sub.1**, **187.sub.2** that are fluidly connected to the manifolds **170.sub.1**, **170.sub.2** respectively. In this embodiment, the control valve **185** is a solenoid valve controllable by the controller **180** and its normal position is that associated with diverting air flow from the compressor **168** to the fluid lines **187.sub.1**, **187.sub.2**. The control valve **185** may be configured differently in other embodiments. As can be seen, the fluid lines **187.sub.1**, **187.sub.2** bypass the solenoid valves **174** of the two subsets **175.sub.1**, **175.sub.2**, and therefore, in the compressor feed mode, the air supplied by the

compressor **168** bypasses the solenoid valves **174** that are connected to the pilot ports of the manifold valves **172.sub.1-172.sub.9**. In the compressor feed mode, a manifold valve **172.sub.1** associated with the given inflatable section of the mattress **100** that is to be pressurized is opened to allow air from the corresponding manifold to enter the manifold valve **172.sub.i**. The air flowing into the manifolds **170.sub.1, 170.sub.2** from the fluid lines **187.sub.1, 187.sub.2** then enters the manifold valve **172.sub.i** that is opened and enters the associated inflatable section. The pressurization of the inflatable section is stopped once the pressure sensor **225** associated with the given inflatable section indicates that the pressure inside the inflatable section has reached the target pressure $P_{sub.x}$. Since the flow rate of the compressor **168** is low, pressure within the given inflatable section increases slowly which may help in adjusting the pressure within the inflatable section more precisely.

[0184] In the compressor feed mode, the manifold valves **172.sub.1-172.sub.9** are piloted by air supplied by the reservoir **230**. Notably, each solenoid valve **174** is supplied air by the reservoir **230**, and the air is transmitted to the pilot ports of the manifold valves **172.sub.1-172.sub.9** to maintain them in a closed position. In order to open the selected manifold valve **172.sub.i**, the associated solenoid valve **174** is actuated to cease air flow from reservoir **230** to the pilot port of the selected manifold valve **172.sub.i**.

[0185] As will be understood, in the compressor feed mode, the blowers **166.sub.1, 166.sub.2** can remain deactivated since they are not used for driving air to the inflatable sections of the mattress **100**. In some cases, at least one of the blowers **166.sub.1, 166.sub.2** could remain activated for example to feed air to the microclimate management layer **200** via the microclimate management valve **222**.

[0186] Implementation of the compressor feed mode reduces the use of the blowers **166.sub.1, 166.sub.2** which can be helpful to limit noise emitted by the pneumatic control assembly **150** since the blowers **166.sub.1, 166.sub.2** are generally louder than the compressor **168**. In particular, an intermittent noise generated by the blowers **166.sub.1, 166.sub.2** caused by turning the blowers **166.sub.1, 166.sub.2** on and off continuously to adjust the pressure of the inflatable sections of the mattress **100** can be avoided or otherwise minimized by the compressor feed mode. As will be appreciated, the reduction in noise can be helpful to reduce discomfort for a patient lying on the mattress **100**. In addition, a risk of overheating the blowers **166.sub.1, 166.sub.2** from excessive use may be reduced by implementing the compressor feed mode. Furthermore, the compressor feed mode may be helpful in high altitude environments such as in a hospital located in a high altitude municipality. Notably, in such an environment, the output pressure of the blowers may be more limited and thus the compressor feed mode could be implemented to compensate for this limitation.

[0187] It is contemplated that the compressor feed mode may not be implemented in other embodiments. In such cases, the air flow provided by the blowers **166.sub.1, 166.sub.2** is driven to the inflatable sections whether the pressure adjustment required at the inflatable sections is small or large. As will be understood, steps **1050** and **1060** of method **1000** and the fluid lines **187.sub.1, 187.sub.2** shown in FIG. **11** that fluidly connect the compressor **168** to the manifold valves **172.sub.1-172.sub.9** may be omitted in such embodiments.

[0188] Returning to FIG. **10**, if at step **1050**, the pressure adjustment for the given inflatable section is determined to be greater than the threshold pressure adjustment $\Delta P_{sub.lim}$, the controller **180** proceeds to step **1070**. At step **1070**, the controller **180** compares the target pressure $P_{sub.x}$ to a threshold pressure $P_{sub.T}$ in order to determine which of the direct feed mode or the dual-stage compression mode the controller **180** should implement to increase the pressure within the given inflatable section. The threshold pressure $P_{sub.T}$ is a predetermined pressure that is less than the pressure capacity of each blower **166.sub.1, 166.sub.2**. In this example, the threshold pressure $P_{sub.T}$ is 2.7 kPa (i.e., half of the pressure capacity of each blower **166.sub.1, 166.sub.2**). If the target pressure $P_{sub.x}$ is not greater than the threshold pressure $P_{sub.T}$, the method proceeds to step **1080** where the direct feed mode is implemented by the controller **180** in order to increase the

pressure of the inflatable section via a single one of the blowers **166.sub.1**, **166.sub.2**. In other words, the direct feed mode is implemented when the target pressure **P.sub.x** is within the range of the pressure capacity of an individual one of the blowers **166.sub.1**, **166.sub.2**. In this example, the value of the threshold pressure **P.sub.T** is less than the pressure capacity of each blower **166.sub.1**, **166.sub.2** in order to ensure that the noise emitted by the blowers **166.sub.1**, **166.sub.2** during operation will not be excessive in the direct feed mode as the blowers generate more noise the closer they get to operating at their maximum pressure capacity. The threshold pressure **P.sub.T** may be a different proportion of the pressure capacity of each blower **166.sub.1**, **166.sub.2** in other embodiments (e.g., 40% of the pressure capacity of each blower).

[0189] In the direct feed mode, a blower **166.sub.1** or **166.sub.2** draws air from a respective inlet of the pneumatic control assembly **150** and drives said air to the corresponding one of the manifolds **170.sub.1**, **170.sub.2**. Within the manifold, the air is then directed towards the manifold valve **172.sub.i** corresponding to the given inflatable section. Thus, in the direct feed mode, the air provided to the given inflatable section of the mattress **100** is compressed a single time by one of the blowers **166.sub.1**, **166.sub.2**. The air discharged to the given inflatable section may thus be pressurized up to the maximum capacity of an individual one of the blowers **166.sub.1**, **166.sub.2**. The direct feed mode may thus also be referred to as a “single-stage compression mode”.

[0190] An example of operation in the direct feed mode will now be described in relation to pressurizing the back support section **136**. With reference to FIG. **11**, in the direct feed mode, the controller **180** opens the first inlet valve **176.sub.1** and the first control valve **179.sub.1** and activates the blower **166.sub.1** to allow external air to be drawn into the pneumatic control assembly **150** via an inlet opening (not shown) and through a corresponding muffler **305**. The muffler **305** is configured to limit noise generated by the air being drawn into the pneumatic control assembly **150**. In the direct feed mode, the interconnection valves **178.sub.1**, **178.sub.2** which are used to selectively interconnect the manifolds **170.sub.1**, **170.sub.2** are closed. The blower **166.sub.1** drives the air drawn from the exterior to the manifold **170.sub.1**. The manifold valve **172.sub.7** is opened by actuating the corresponding solenoid valve **174** and thus the air is directed to the back support section **136** that is in selective communication with the manifold **170.sub.1** via the manifold valve **172.sub.7**.

[0191] As will be appreciated, a similar operation may be performed to allow air flow from the blower **166.sub.1** to the other air-powered parts of the mattress **100** that are associated with the other manifold valves **172.sub.2-172.sub.9**.

[0192] Although the operation of the direct feed mode has been described in relation to the back support section **136**, it will be appreciated that the same procedure is executed for increasing the pressure of the other inflatable sections associated with the manifold **170.sub.1**. Moreover, as mentioned above, multiple inflatable sections that are fluidly connected to the manifold **170.sub.1** may be inflated at the same time by opening the corresponding manifold valves **172.sub.1-172.sub.9** of the manifold **170.sub.1**.

[0193] Furthermore, the reader will understand that the blower **166.sub.2** may similarly distribute air in the direct feed mode to other air-powered parts of the mattress **100** that are associated with the manifold valves **172.sub.1-172.sub.9** of the second manifold **170.sub.2**. The operation of the direct feed mode in relation to the inflatable sections associated with the blower **166.sub.2** will therefore not be described herein. Elements described above with regard to the functionality of the blower **166.sub.1** (e.g., inlet valve **176.sub.1**) have been identified with similar reference numbers for the blower **166.sub.2** (e.g., inlet valve **176.sub.2**), with a subscript “2” instead.

[0194] Returning to FIG. **10**, if the target pressure **P.sub.x** is greater than the threshold pressure **P.sub.T**, the method proceeds to step **1090** where the controller **180** implements the dual-stage compression mode in order to use both blowers **166.sub.1**, **166.sub.2** to increase the pressure of the given inflatable section of the mattress **100**. In other words, when the target pressure **P.sub.x** exceeds the pressure capacity of a single blower **166.sub.1**, **166.sub.2**, the dual-stage compression

mode is engaged whereby the blowers **166.sub.1**, **166.sub.2** are fluidly connected in series to enable two stages of compression by the blowers **166.sub.1**, **166.sub.2**.

[0195] The dual-stage compression mode will now be described in greater detail with reference to FIG. **11**. In order to fluidly connect the two blowers **166.sub.1**, **166.sub.2**, one of the interconnection valves **178.sub.1**, **178.sub.2** is opened to fluidly connect the outlet of one blower with the inlet of the other blower. The selection of which interconnection valve **178.sub.1**, **178.sub.2** to open depends on the inflatable section that is going to be pressurized. If the inflatable section to be pressurized is associated with one of the manifold valves **172.sub.1-172.sub.9** of the first manifold **170.sub.1**, then the first interconnection valve **178.sub.1** is opened and the second interconnection valve **178.sub.2** is closed so that the first blower **166.sub.1** can ultimately supply air to the first manifold **170.sub.1** (i.e., the first blower **166.sub.1** provides the last stage of compression after the second blower **166.sub.2**). Notably, if the first interconnection valve **178.sub.1** is open, then the outlet of the second blower **166.sub.2** is in communication with the inlet of the first blower **166.sub.1** (with the first control valve **179.sub.1** being opened). If, on the other hand, the inflatable section to be pressurized is associated with one of the manifold valves **172.sub.1-172.sub.9** of the second manifold **170.sub.2**, then the second interconnection valve **178.sub.2** is opened and the first interconnection valve **178.sub.1** is closed so that the second blower **166.sub.2** can ultimately supply air to the second manifold **170.sub.2**. As will be appreciated, if the second interconnection valve **178.sub.2** is open (with the second control valve **179.sub.2** being opened), then the outlet of the first blower **166.sub.1** is in communication with the inlet of the second blower **166.sub.2**.

[0196] Thus, if for example, the back support section **136** associated with the manifold valve **172.sub.7** of the first manifold **170.sub.1** is to be pressurized in the dual-stage compression mode, the manifold valve **172.sub.7** is opened, the first inlet valve **176.sub.1** is closed (to disconnect the first blower **166.sub.1** from the corresponding inlet opening), the first interconnection valve **178.sub.1** and the first control valve **179.sub.1** are opened, the second interconnection valve **178.sub.2** is closed, and the second inlet valve **176.sub.2** and second control valve **179.sub.2** are opened (to fluidly connect the second blower **166.sub.2** to the corresponding inlet opening). The manifold valves **172.sub.1-172.sub.9** of the second manifold **170.sub.2** are closed so that air flow from the second blower **166.sub.2** is directed to the first blower **166.sub.1**. The second blower **166.sub.2** is activated and thus the second blower **166.sub.2** draws air through the inlet valve **176.sub.2** and drives the air to the first blower **166.sub.1**, where the air is compressed again by the first blower **166.sub.1**. The first blower **166.sub.1** thus drives the air to the open manifold valve **172.sub.7** and thus to the back support section **136**. As will be appreciated, the air that is fed to the back support section **136** is compressed two times, namely one time by each blower **166.sub.1**, **166.sub.2**.

[0197] As will be appreciated, more than one inflatable section of the mattress **100** may be inflated at the same time in the dual-stage compression mode. For instance, multiple manifold valves of the same manifold could be opened to feed air to their associated inflatable sections that has been compressed by both blowers **166.sub.1**, **166.sub.2**.

[0198] It is contemplated that in other embodiments in which the dual compression provided by the blowers **166.sub.1**, **166.sub.2** is not deemed necessary (i.e., the target pressure **P.sub.x** is not expected to exceed the capacity of a single blower), the pneumatic control assembly **150** could include a single blower. For example, the single blower could be used to pressurize and depressurize all of the inflatable sections of the mattress **100**.

[0199] Conversely, it is also contemplated that, in some embodiments, the direct feed mode may be omitted such that the controller **180** operates the pneumatic control assembly **150** in the dual-stage compression mode in all cases in which one of the inflatable sections is to be pressurized (unless the compressor feed mode is also implemented as described above). This may be done for example to reduce the noise emitted by the blowers **166.sub.1**, **166.sub.2** since they tend to generate more

noise when they operate closer to their maximum pressure capacity.

[0200] It is to be understood that the method **1000** is just an example of a process for determining which operation mode the controller **180** should implement, and that the method **1000** could be formulated differently to achieve a similar result. As an example, the method **1000** may instead calculate a difference between the target pressure $P_{sub.T}$ and the current pressure $P_{sub.c}$, and categorize the result to determine the operation mode to implement. Moreover, the order of the steps of the method **1000** may be different in other embodiments. Furthermore, while the operation modes have been generally described in relation to one inflatable section having to undergo a pressure adjustment, the method **1000** and the operation modes can be executed simultaneously for multiple inflatable sections of the mattress **100**.

[0201] The construction of the manifolds **170.sub.1**, **170.sub.2** will now be described in greater detail with reference to FIGS. **12** to **14**. In this embodiment, the manifolds **170.sub.1**, **170.sub.2** are identical to one another and thus a single one of the manifolds **170.sub.1**, **170.sub.2** (which will be referred to as manifold **170.sub.1**) will be described herein. The manifold **170**; has a main body **190** that receives respective ones of the manifold valves **172.sub.1-172.sub.9** therein. The main body **190** also receives a corresponding inlet valve **176.sub.i**, interconnection valve **178.sub.i** and control valve **179.sub.1**. The main body **190** has an inner lateral end **202** and an outer lateral end **204** opposite the inner lateral end **202**. The inner and outer lateral ends **202**, **204** correspond to the inner and outer lateral ends of the manifold **170.sub.i**. In use, as shown in FIG. **8**, the outer lateral end **204** is disposed near the right end **164** of the housing **152** while the inner lateral end **202** is disposed near a centerline of the housing **152** that bisects a width of the housing **152**. Furthermore, in use, the first and second manifolds **170.sub.1**, **170.sub.2** are oriented such that their inner lateral ends **202** are closer to each other than the outer lateral ends **204**.

[0202] An upper cover **192** of the manifold **170.sub.i** is secured to the main body **190** by multiple fasteners **194**. As best shown in FIG. **14**, the upper cover **192** defines a plurality of circular openings **195** through which part of each manifold valve **172.sub.1-172.sub.9**, and each valve **176.sub.1**, **178.sub.1**, **179.sub.1** protrudes. A gasket **197** is disposed between the upper cover **192** and the main body **190** to provide a seal along an interface between the cover **192** and the main body **190**, as well as between the upper cover **192** and the valves **172.sub.1-172.sub.9**, **176.sub.1**, **178.sub.1**, **179.sub.1**. The gasket **197** defines openings corresponding to the openings **195** of the upper cover **192**.

[0203] A lower cover **265** is secured to an underside of the main body **190** and defines respective openings **237**, **239**, **241** for the outlets of each of the valves **176.sub.i**, **178.sub.i** and **179.sub.i** to protrude therefrom.

[0204] The main body **190** is shown in greater detail in FIGS. **16** and **17**. As can be seen, the main body **190** defines a main valve chamber **212** which receives the manifold valves **172.sub.1-172.sub.9**. The main body **190** also defines a secondary valve chamber **214** that is separated from the main chamber **212** by a dividing wall **215**. The secondary valve chamber **214** is disposed adjacent to the outer lateral end **204**. The secondary valve chamber **214** receives the inlet valve **176.sub.i**, the interconnection valve **178.sub.i** and the control valve **179.sub.1**. The main and secondary valve chambers **212**, **214** are defined on an upper side of the main body **190** and are closed off by the upper cover **192** and the gasket **197**.

[0205] The main body **190** has a bottom wall **216** that defines a plurality of openings, including valve openings **218** opening into the main chamber **212** and valve openings **221** opening into the secondary valve chamber **214**. As will be explained in greater detail further below, the valve openings **218** are shaped to help orient the insertion of corresponding connectors **223** (FIG. **14**). In particular, each valve opening **218** has a central circular portion and two wing portions extending from the circular portion. Furthermore, as shown in FIG. **17**, the bottom wall **216** also defines an inner connecting aperture **232** near the inner lateral end **202** that opens into the main chamber **212** of the manifold **170.sub.i**, and two outer connecting apertures **234**, **235** that are closer to the outer

lateral end **204** (near the dividing wall **215**) that also open into the main chamber **212**. The connecting apertures **234**, **235** are distanced from each other in the longitudinal direction of the manifold **170.sub.i**. The connecting apertures **232**, **234**, **235** are provided for fluidly connecting the main chamber **212** to other parts of the manifold **170.sub.i** (e.g., the secondary valve chamber **214**) or to other parts of the pneumatic control assembly **150**, as will be described in greater detail below.

[0206] The connecting apertures **234**, **235** are aligned with respective openings of the lower cover **265**.

[0207] As shown in FIG. **16**, valve supports **224** protrude from the bottom wall **216** for supporting the manifold valves **172.sub.1-172.sub.9** and the valves **176.sub.i**, **178.sub.i**, **179.sub.1**. Three valve supports **224** are disposed around each valve opening **218**, **221** for supporting a corresponding valve.

[0208] It is contemplated that, in some embodiments, the manifolds **170.sub.1**, **170.sub.2** could share a common upper cover **192** as shown in the alternative embodiment illustrated in FIG. **28**. In particular, as shown in FIG. **28**, the common upper cover **192** is affixed to the main body **190** of each of the manifolds **170.sub.1**, **170.sub.2** so as to close off the main chamber **212** and the secondary valve chamber **214** of each of the manifolds **170.sub.1**, **170.sub.2**. Similarly, the underlying gasket **197** (not shown in FIG. **28**) extends across both manifolds **170.sub.1**, **170.sub.2** to form a common gasket. This configuration may provide an improved seal along the top of the manifolds **170.sub.1**, **170.sub.2**. The lower cover **265** could similarly be a common lower cover that is fixed to the main bodies **190** of both manifolds **170.sub.1**, **170.sub.2**.

[0209] As best shown in FIG. **17**, on the underside of the main body **190**, the main body **190** defines an interconnection channel **226** extending generally in a lateral direction of the manifold **170.sub.i**. In use, the interconnection channel **226** is covered by a cover **228** (FIG. **13**) that is fastened to the main body **190**. As shown in FIG. **14**, a gasket **243** is disposed between the cover **228** and the main body **190** to provide a seal therebetween. The cover **228** defines a connecting aperture **236** near the inner lateral end **202** of the main body **190** for providing access to the interconnection channel **226**. The lower cover **265** also defines an aperture **267** (FIGS. **13**, **14**) aligned with the interconnection channel **226**, at an end of the interconnection channel **226** opposite the connecting aperture **236**. The interconnection channel **226** is used for fluidly connecting the two manifolds **170.sub.1**, **170.sub.2**. In particular, the interconnection channel **226** of the first manifold **170.sub.1** is fluidly connected to the main chamber **212** of the second manifold **170.sub.2** while the interconnection channel **226** of the second manifold **170.sub.2** is fluidly connected to the main chamber **212** of the first manifold **170.sub.1**. To that end, the connecting aperture **236** of the first manifold **170.sub.1** is fluidly connected to the inner connecting aperture **232** of the second manifold **170.sub.2**, while the connecting aperture **236** of the second manifold **170.sub.2** is fluidly connected to the inner connecting aperture **232** of the first manifold **170.sub.1**. Notably, with reference to FIG. **9**, in this embodiment, the fluid connection between the connecting apertures **232**, **236** of the first and second manifolds **170.sub.1**, **170.sub.2** is provided by channels **238** defined by the base portion **154** of the housing **152** of the pneumatic control assembly **150**. The channels **238** are covered by a lid (not shown) fixed to the underside of the base portion **154** of the housing **152**.

[0210] The outer connecting apertures **234**, **235** are provided for fluidly connecting the main chamber **212** of the manifold **170.sub.i** with the corresponding blower **166.sub.1**. Notably, the outer connecting aperture **234**, **235** that is closest to the blower **166.sub.i** corresponding to that manifold **170.sub.i** (i.e., that feeds that manifold **170**; in the direct feed mode) is used for fluidly connecting the main chamber **212** with the corresponding blower **166.sub.1**. An internal channel (not shown) defined by the base portion **154** of the housing **152** establishes the fluid connection between the main chamber **212** (via the corresponding outer connecting aperture **234**, **235**) and the blower chamber **187** (FIG. **8**) in which the corresponding blower **166.sub.1** is disposed. The other outer connecting aperture **234**, **235** (i.e., the one that is furthest from the corresponding blower

166.sub.1) is provided only so that the same manifold structure can be used for both manifolds **170.sub.1**, **170.sub.2** since the manifolds **170.sub.1**, **170.sub.2** are oriented such that they are rotationally offset by 180 relative to each other.

[0211] With reference to FIG. **14**, in this embodiment, each manifold valve **172.sub.1-172.sub.9** includes a valve module **210** that is received within the main chamber **212** defined by the main body **190**, and an associated connector **223** that is fixed to the main body **190** and protrudes from the underside thereof. The valve module **210** is actuatable to selectively fluidly connect the main chamber **212** with the air-powered part to which the associated connector **223** is connected.

[0212] As shown in FIGS. **18** and **19**, in this example, each connector **223** includes a first connector portion **240** and a second connector portion **242** (shown disassembled from each other in FIG. **14**). The first connector portion **240** has a cylindrical body **244** and a flange **246** extending around a midsection of the cylindrical body **244**. An upper end of the cylindrical body **244** defines an opening **245** which, as will be explained in more detail below, is selectively blocked by the corresponding manifold valve **172.sub.i** to allow or prevent air flow therethrough. Two wings **248** extend from the cylindrical body **244** opposite from each other, on an upper side of the flange **246**. Each wing **248** is curved around a same axis as the cylindrical body **244** and has a horizontal portion **250** and an abutting portion **252**. The horizontal portion **250** is spaced from the flange **246** and extends generally parallel to the flange **246**. The abutting portion **252** extends vertically downwardly from the horizontal portion **250** and is connected to the flange **246**. The horizontal portion **250** may have a protruding ridge **254** extending from a lower surface of the horizontal portion **250**. An O-ring **256** is disposed on the upper side of the flange **246**, around a circumferential edge thereof, to provide a seal against the main body **190** of the manifold **170.sub.1**.

[0213] The second connector portion **242** of each connector **223** is a tubular elbow having a vertical portion **258** and a horizontal portion **260**. A flange **262** extends circumferentially around the upper end of the vertical portion **258**. The flange **262** is connected (e.g., welded or mechanically fastened) to the flange **246** of the first connector portion **240**. As shown in FIG. **15**, part of the cylindrical body **244** of the first connector portion **240** is received into the vertical portion **258** of the second connector portion **242**. In use, the horizontal portion **260** is engaged by pneumatic tubing (not shown) that fluidly connects the connector **223** to the associated air-powered part of the mattress **100**. For example, part of the pneumatic tubing may surround part of the horizontal portion **260** and be snugly fitted therewith. As shown in FIG. **18**, the horizontal portion **260** has an end that defines an opening **264** through which air is discharged to the associated air-powered part of the mattress **100** (e.g., an inflatable section or the microclimate management layer **200**). Thus, air that enters the opening **245** of the first connector portion **240** is discharged via the opening **264** of the second connector portion **242**.

[0214] It is contemplated that, in some embodiments, the second connector portion **242** may be omitted. In such cases, the pneumatic tubing could be directly connected to the first connector portion **240** for example.

[0215] With reference to FIG. **15**, each connector **223** is fixed to the main body **190** by aligning the wings **248** with the wing portions of a corresponding valve opening **218** and inserting the cylindrical body **244** therein such that the cylindrical body **244** protrudes into the main chamber **212** of the main body **190**. The connector **223** is then rotated counterclockwise until the bottom wall **216** of the main body **190** contacts the abutting portions **252** of the wings **248**. The protruding ridges **254** of the horizontal portions **250** of the wings **248** may help secure the connector **223** in place. The O-ring **256** provides a seal along the interface of the connector **223** with the main body **190**, namely along the lower surface of the main body **190**.

[0216] With reference to FIGS. **20** to **22**, the valve module **210** of each manifold valve **172.sub.i** includes an upper member **270**, a lower member **272** and a diaphragm **274** disposed therebetween. The upper member **270** has a central portion **276** and an elbow connector **278** extending upwardly

from the central portion **276**. The elbow connector **278** defines a pilot port **280** of the valve module **210** for piloting the respective manifold valve **172.sub.i**. In use, pneumatic tubing is connected to the elbow connector **278** to fluidly connect a corresponding solenoid valve **174** thereto as explained above and as shown schematically in FIG. **11**. The upper member **270** also has a peripheral annular portion **282** extending radially outwardly from the central portion **276**. A plurality of openings **284** are defined between the central portion **276** and the peripheral annular portion **282**. As will be explained below, the openings **284** are used for interlockingly secure the upper member **270** to the lower member **272**.

[0217] The lower member **272** is generally annular and has a central rim portion **286** and an outer peripheral portion **288** extending outwardly therefrom. The central rim portion **286** defines a central aperture **290** (FIG. **22**) and has an upwardly extending annular ridge **291** (FIG. **21**). A plurality of detents **292** extend upwardly from the outer peripheral portion **288** to engage the upper member **270**. In particular, the detents **292** are inserted into corresponding ones of the openings **284** of the upper member **270** to retain the upper and lower members **270**, **272** together. To that end, each detent **292** has a protruding edge **294** that, when the detent **292** is inserted into the corresponding opening **284**, engages an upper surface of the peripheral annular portion **282** of the upper member **270**. The detents **292** are thus prevented from disengaging the upper member **270** unless the detents **292** are forcibly pushed inwardly so that the detents **292** can be removed from openings **284**.

[0218] With reference to FIGS. **21** and **22**, the diaphragm **274** of each valve module **210** has an outer peripheral portion **296** and a central portion **298** that are connected by a wall portion **301** extending therebetween. The wall portion **301** extends vertically between the outer peripheral portion **296** and the central portion **298**. In particular, the wall portion **301** has a first end **315** (which can be referred to as a “lower end” in this embodiment) that is connected to the central portion **298**, and a second end **317** (which can be referred to as an “upper end” in this embodiment) that is connected to the outer peripheral portion **296**. The diaphragm **274** is generally cup-shaped as the outer peripheral portion **296** and the central portion **298** are disposed at different heights. In particular, as can be seen in FIG. **21**, in this embodiment, a cross-sectional profile of the diaphragm **274** is generally W-shaped, namely as the central portion **298** is vertically closer to the outer peripheral portion **296** than the first (lower) end of the wall portion **301** (i.e., the height between the outer peripheral portion **296** and the first (lower) end of the wall portion **301** is greater than the height between the outer peripheral portion **296** and the central portion **298**). This shape may be helpful to prevent or otherwise minimize a permanent deformation of the diaphragm **274** during use, which may be more likely for example if the central portion **298** were at the same vertical height as the lower end of the wall portion **301**. The diaphragm **274** is made of a pliable material (e.g., silicone) that is able to deform and bias back to its original shape. The diaphragm **274** may be made by molding, such as by injection molding or transfer molding.

[0219] The diaphragm **274** is fixed between the upper and lower members **270**, **272** by positioning its central portion **298** into the central aperture **290** of the lower member **272** such that the outer peripheral portion **296** sits on the lower member **272**. The upper member **270** is then engaged with the lower member **272** as mentioned above via the detents **292** and openings **284**. Once the upper member **270** is engaged with the lower member **272**, the diaphragm **274** is fixed therebetween and forms, together with the upper member **270**, an inner chamber **303** (FIG. **21**) of the valve module **210**. The inner chamber **303** is accessible via the pilot port **280**. As can be seen in FIG. **21**, part of the diaphragm **274** is compressed between the annular ridge **291** of the lower member **272** and a lower annular ridge **285** of the upper member **270**. Moreover, the outer peripheral portion **296** is compressed and provides a seal for the inner chamber **303** of the valve module **210**. As will be described with respect to another embodiment further below, the diaphragm **274** could be set in a different orientation than that described above.

[0220] With reference to FIG. **15**, each valve module **210** is seated on three corresponding valve

supports **224** surrounding one of the valve openings **218**. The lower member **272** of each valve module **210** has recesses **277** (one of which is shown in FIG. **21**) in which the upper ends of the valve supports **224** are inserted. The valve supports **224** are thus used as positioning features for the valve module **210**. Once the valve module **210** is seated on the valve supports **224**, the valve module **210** is aligned with the corresponding connector **223**. More specifically, the diaphragm **274** is generally concentric with the cylindrical body **244** of the outlet connector **223**. When air is not supplied to the pilot port **280** of the valve module **210**, the diaphragm **274** is in the position shown in solid lines in FIG. **15**, corresponding to the open position of the manifold valve **172.sub.i**, whereby the main chamber **212** of the manifold **170.sub.i** is in fluid connection with the connector **223**. The diaphragm **274** is therefore biased by default to the open position of the manifold valve **172.sub.i**. Air can thus enter and exit the main chamber **212** of the manifold **170.sub.i** via the open manifold valve **172.sub.i**. Conversely, when air is supplied to the pilot port **280**, the diaphragm **274** deforms to the position shown in dashed lines in FIG. **15** (corresponding to the closed position of the manifold valve **172.sub.i**), in which the diaphragm **274** blocks the opening **245** of the connector **223**, thereby fluidly disconnecting the main chamber **212** of the manifold **170.sub.i** from the connector **223**. As mentioned above, in this embodiment, the manifold valve **172.sub.i** is normally in the closed position, and when air flow to the pilot port **280** thereof ceases (due to actuation of the corresponding solenoid valve **174**), the manifold valve **172.sub.i** moves to the open position. It is contemplated that, in other embodiments, the diaphragm **274** could be biased to be in the closed position shown in dashed lines in FIG. **15** in other embodiments. For instance, in such cases, instead of feeding air into the pilot port **280** of the valve module **210** to block air flow to the connector **223**, air may be vacuumed from the pilot port **280** to place the diaphragm **274** in the open position of the manifold valve **172.sub.i**, thereby allowing air flow through the connector **223**. [0221] The manifold valves **172.sub.1-172.sub.9** may be relatively quiet in operation because they integrate a diaphragm. Notably, operation of the manifold valves may be quieter than that of solenoid valves that are typically used for controlling air flow in inflatable mattresses. Moreover, the modular design of the manifold valves **172.sub.1-172.sub.9** may make the assembly of the manifold **170.sub.i** relatively simple.

[0222] It is contemplated that the manifold valves **172.sub.1-172.sub.9** could be configured differently in other embodiments. For instance, in some cases, the relationship between the upper and lower members **270**, **272** of the valve module **210** may be inversed such that the upper member **270** has the detents and the lower member **272** has the corresponding openings for receiving the detents.

[0223] For instance, FIGS. **24** to **26** illustrate a valve module **210'** according to an alternative embodiment. The valve module **210'** has an upper member **270'**, a lower member **272'** and the diaphragm **274** disposed therebetween. The upper member **270'** has a central portion **276'** and a peripheral annular portion **282'** extending radially outwardly from the central portion **276'**. An elbow connector **278'** extends upwardly from the central portion **276'** and defines a pilot port **280'** of the valve module **210'** for piloting the respective manifold valve **172.sub.i**. The upper member **270'** is generally-cup shaped as an upper end of the central portion **276'** being disposed at a different height from the peripheral annular portion **282'**. In this embodiment, the peripheral annular portion **282'** includes tongues **284'** disposed at a radial edge of the peripheral annular portion **282'** and which are used for interlockingly securing the upper member **270'** to the lower member **272'**. In this example, five tongues **284'** are provided, spaced equally circumferentially about the peripheral annular portion **282'**.

[0224] The lower member **272'** is generally annular and has a central rim portion **286'** and an outer peripheral portion **288'** extending outwardly therefrom. As best shown in FIG. **25**, the central rim portion **286'** defines a central aperture **290'** and has an upwardly extending annular ridge **291'**. A plurality of detents **292'** extend upwardly from the outer peripheral portion **288'** to engage the upper member **270'**. In particular, each detent **292'** defines an opening **293'** for receiving a

corresponding one of the tongues **284'** of the upper member **270'**. Notably, once the detents **292'** engage the tongues **284'** by inserting the tongues **284'** into the openings **293'**, the upper and lower members **270'**, **272'** are secured to each other. The detents **292'** are prevented from disengaging the upper member **270'** unless the detents **292'** are forcibly pushed outwardly so that the tongues **284'** can be removed from openings **293'**.

[0225] As shown in FIGS. **25** and **26**, in this alternative embodiment, the diaphragm **274** is set in an opposite orientation to that described in the previous embodiment. Notably, in this embodiment, the central portion **298** is disposed vertically higher than the outer peripheral portion **296**. Thus, the inner chamber **303** of the valve module **210'** is defined in part by an opposite side of the central portion **298** than in the previously described embodiment. This orientation of the diaphragm **274** may minimize stresses generated along the central portion **298** after repeated deformations and thus improve durability.

[0226] The diaphragm **274** is fixed between the upper and lower members **270'**, **272'** by positioning its central portion **298** into the central aperture **290'** of the lower member **272'** such that the outer peripheral portion **296'** sits on the lower member **272'**. The upper member **270'** is then engaged with the lower member **272'** as mentioned above via the detents **292'** and tongues **284'**. Once the upper member **270'** is engaged with the lower member **272'**, the diaphragm **274** is fixed therebetween and forms, together with the upper member **270'**, the inner chamber **303** of the valve module **210'**. The inner chamber **303** is accessible via the pilot port **280''**. As can be seen in FIG. **25**, part of the diaphragm **274** is compressed between the annular ridge **291'** of the lower member **272'** and a lower annular ridge **285'** of the upper member **270'**. Moreover, the outer peripheral portion **296** of the diaphragm **274** is compressed and provides a seal for the inner chamber **303** of the valve module **210'**.

[0227] The inlet valve **176.sub.i**, the interconnection valve **178.sub.i** and the control valve **179.sub.1** have a similar configuration to the manifold valves **172.sub.1-172.sub.9**, notably each having the same valve module **210** or **210'**. However, the valves **176.sub.i**, **178.sub.i**, **179**; do not have connectors **223** in this embodiment. Rather, as shown in FIG. **16**, the main body **190** has three annular protrusions **217** extending upwardly from the bottom wall **216** in the secondary valve chamber **214**, each annular protrusion **217** defining a corresponding valve opening **221**. The valve opening **221** defined by each annular protrusion **217** is selectively blocked by the diaphragm **274** of the corresponding valve module **210** as described above with respect to the manifold valves **172.sub.1-172.sub.9**. The valve opening **221** defined by each annular protrusion **217** has greater dimensions than the opening **245** of each connector **223** which may allow a greater air flow rate to flow through the valves **176.sub.1**, **178.sub.1**, **179.sub.1**. For instance, an air flow rate through the valves **176.sub.1**, **178.sub.1**, **179.sub.1** may be at least 50% greater than the air flow rate through the manifold valves **172.sub.1-172.sub.9**. In this example, the air flow rate through the valves **176.sub.i**, **178.sub.i**, **179.sub.i** is approximately 100% greater than the air flow rate through the manifold valves **172.sub.1-172.sub.9**.

[0228] Although the manifolds **170.sub.1**, **170.sub.2** are identical to each other in this embodiment, which may be beneficial for simplifying manufacturing of the pneumatic control assembly **150**, it is contemplated that the manifolds **170.sub.1**, **170.sub.2** could be configured differently from each other in other embodiments.

[0229] The pneumatic system of the pneumatic control assembly **150** may be configured differently in other embodiments. For instance, certain components (e.g., one or more inflatable sections of the mattress **100**) of the pneumatic system may be omitted and/or additional components may be provided. As an example, the pneumatic diagram of an alternative embodiment of the pneumatic system is illustrated in FIG. **27**. In this embodiment, the compressor **168** cannot itself feed air to the various inflatable parts of the mattress **100** since the above-described fluid lines **187.sub.1**, **187.sub.2** are omitted such that the compressor feed mode described above cannot be implemented. Rather, it is one or both blowers **166.sub.1**, **166.sub.2** that drive air into or pull air from the air-

powered parts of the mattress **100** in the manner already described above. Similarly, some of the diaphragm valves (and their associated solenoid valves) described in the previous embodiment are omitted to reduce redundancy. For instance, in this embodiment, the purge valve **220** and the control valves **179.sub.1**, **179.sub.2** are omitted. Furthermore, in this embodiment, the foot support section **144** of the mattress **100** is omitted and thus the associated valves are also omitted.

[0230] With continued reference to FIG. **27**, the diaphragm valves are also distributed differently in this embodiment. For instance, each manifold **170.sub.1**, **170.sub.2** includes a respective microclimate management valve **222** (fluidly connected to the microclimate management layer **200**) that is piloted by a respective solenoid valve **205**. The microclimate management valves **222** can be used for exhausting air from the mattress **100**. It is to be noted, that some of the inflatable sections of the mattress **100** may be pressurized by air flow directed thereto by a solenoid valve, such as for example the dorsal push bladder **145** as shown in FIG. **27**. Furthermore, in this embodiment, two reservoirs **231** are fluidly connected to the compressor **168** so as to be filled thereby. The reservoirs **231** are in turn fluidly connected to the various solenoid valves of the subsets **175.sub.1**, **175.sub.2**. The reservoirs **231** are provided to compensate for micro air leaks that may occur in the pneumatic system, particularly in situations in which the mattress **100** is not electrically connected to an external power source (e.g., an AC circuit). This may help hold the pressurization of the various bladders of the mattress **100** when the bed **300** is being moved around a hospital for example. In addition, the reservoirs **231** may minimize the number of times the compressor **168** has to be turned on and off to selectively open the diaphragm valves of the manifolds **170.sub.1**, **170.sub.2** which can be helpful to limit the noise produced by operation of the pneumatic control assembly **150**. Check valves **227** are provided to ensure unidirectional flow of air in parts of the pneumatic system and to function as a fail-safe in case of failure of the solenoid valves that might otherwise cause depressurization of the bladders of the mattress **100**. Lastly, in this embodiment, the mattress **100** includes a bottoming-out bladder **255** that is closed and is disposed underneath one or more of the inflatable bladders of the mattress **100**. The pressure of the bottoming-out bladder **255** is communicated to the controller **180** which can determine, based on the pressure readings acquired from the bottoming-out bladder **255**, whether the patient on the mattress **100** is close to being in a “bottoming-out” position whereby the patient is too deeply immersed in the mattress **100** and is thus close to a rigid surface of the mattress **100**. This may allow the controller **180** to adjust the pressure in the inflatable sections of the mattress **100** to counteract the bottoming-out position. The pneumatic system shown in FIG. **27** otherwise functions in the same manner as described in the previous embodiment and therefore will not be described again herein.

[0231] It is contemplated that some of the components of the pneumatic control assembly **150** may be omitted in some embodiments. For instance, in some embodiments, the controller **180** could be disposed externally of the pneumatic control assembly **150**, integrated as part of the bed **300** or within a separate control unit that is supported by the bed **300** (e.g., at a foot end of the bed **300**). In such cases, appropriate connectors are provided to connect the pneumatic control assembly **150** to the external controller.

[0232] Modifications and improvements to the above-described embodiments of the present technology may become apparent to those skilled in the art. The foregoing description is intended to be exemplary rather than limiting. The scope of the present technology is therefore intended to be limited solely by the scope of the appended claims.

Claims

1. A pneumatic control assembly for a patient support surface, comprising: a first air supply device providing air at a high flow rate and a low pressure; a second air supply device providing air at a low flow rate and a high pressure; and at least one first valve configured to be fluidly connected to

- at least one inflatable section of the patient support surface, the at least one first valve being selectively actuated to fluidly connect the first air supply device with a corresponding one of the at least one inflatable section, the at least one first valve being piloted by the second air supply device.
2. The pneumatic control assembly of claim 1, wherein: the first air supply device provides air at a first flow rate and a first pressure; the second air supply device provides air at a second flow rate and a second pressure; the first flow rate is greater than the second flow rate; and the second pressure is greater than the first pressure.
 3. The pneumatic control assembly of claim 1, wherein the first air supply device is a blower and the second air supply device is a compressor.
 4. The pneumatic control assembly of claim 1, wherein the first air supply device has a pressure capacity that is less than 15 kPa.
 5. The pneumatic control assembly of claim 1, wherein the second air supply device has a pressure capacity that is more than 50 kPa.
 6. The pneumatic control assembly of claim 1, wherein: each of the at least one first valve has a pilot port; the pneumatic control assembly further comprises at least one solenoid valve, the at least one solenoid valve being fluidly connected to the second air supply device and to the pilot port of the at least one first valve; and the at least one solenoid valve is selectively actuated to control air flow from the second air supply device to the pilot port of the at least one first valve to thereby actuate the at least one first valve.
 7. The pneumatic control assembly of claim 6, wherein each of the at least one first valve is a diaphragm valve.
 8. The pneumatic control assembly of claim 7, wherein each diaphragm valve includes a diaphragm that is selectively deformable from a first position to a second position in response to the pilot port being in fluid communication with the second air supply device.
 9. The pneumatic control assembly of claim 7, further comprising a manifold comprising the at least one first valve, the manifold being in fluid communication with the first air supply device.
 10. The pneumatic control assembly of claim 1, wherein: the at least one inflatable section is at least one first inflatable section; the first air supply device is in selective fluid communication with the at least one first inflatable section via at least one first valve; the pneumatic control assembly further comprises: a third air supply device providing air at the high flow rate and the low pressure; and at least one second valve configured to be fluidly connected to at least one second inflatable section of the patient support surface, the at least one second valve being selectively actuated to fluidly connect the third air supply device with the at least one second inflatable section, the at least one second valve being piloted by the second air supply device.
 11. The pneumatic control assembly of claim 10, wherein the third air supply device is a blower.
 12. The pneumatic control assembly of claim 1, further comprising: a reservoir that is fluidly connected to the second air supply device, the reservoir being filled by the second air supply device to contain pressurized air therein, the at least one first valve being selectively piloted by the reservoir.
 13. The pneumatic control assembly of claim 1, further comprising a housing enclosing the first air supply device, the second air supply device and the at least one first valve.
 14. A patient support surface for a patient support apparatus comprising: a plurality of inflatable sections; the pneumatic control assembly of claim 1 for controlling air flow to the inflatable sections; and a cover enclosing the inflatable sections and the pneumatic control assembly.
 15. The patient support surface of claim 14, wherein the pneumatic control assembly is located near a foot end of the patient support surface.
 16. The patient support surface of claim 15, wherein the inflatable sections include an inflatable foot section, the pneumatic control assembly being disposed underneath the inflatable foot section.
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